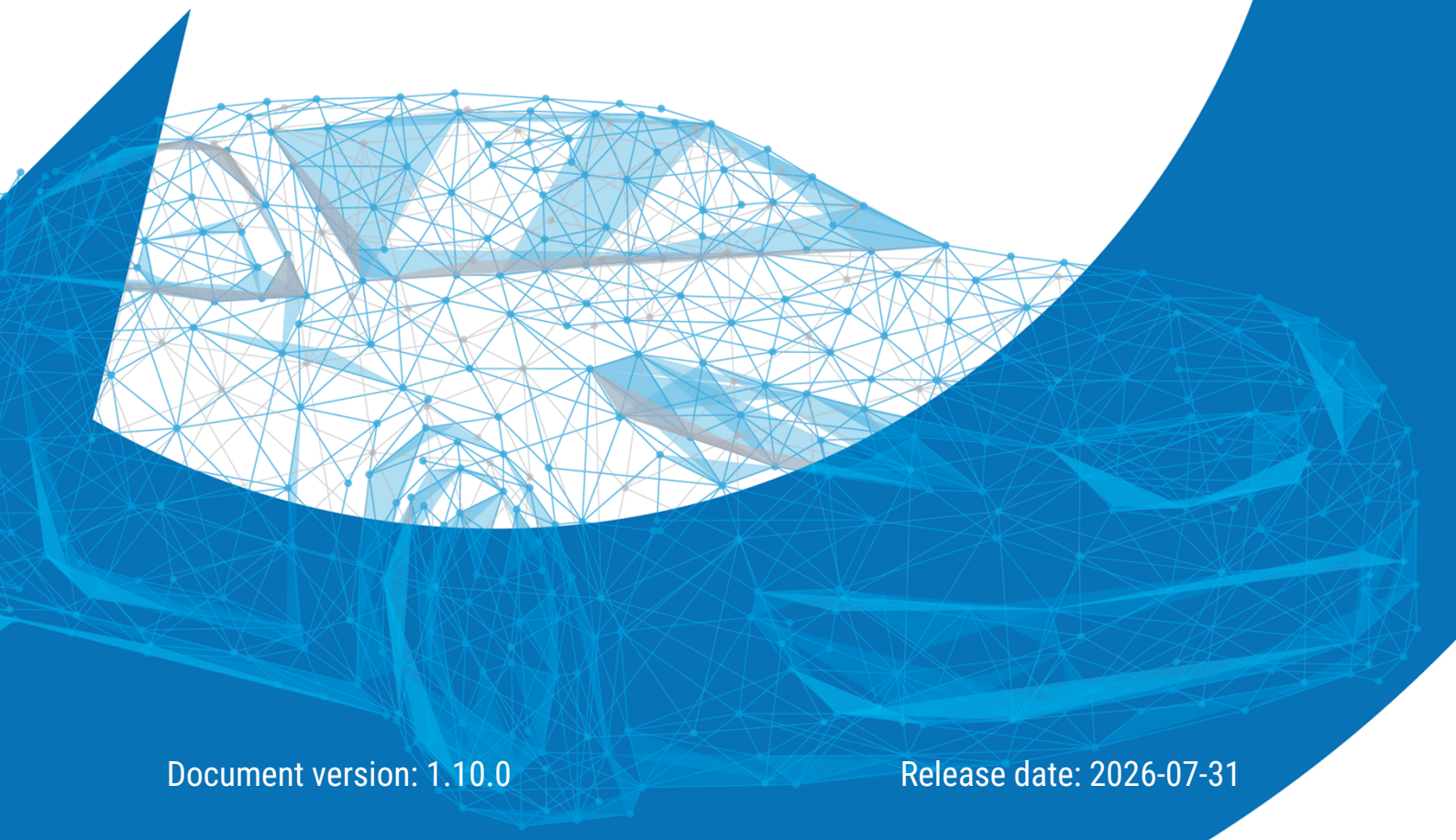
technica engineering

Member of KPIT Group

#OneStepAhead

TECMP

Protocol Specification



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1 General Information

1.1 Introduction

The following protocol description is used by the Capture Modules (CMs) from Technica Engineering. The CM is currently available in five different optimized products for which dedicated user manuals are also available:

- CM Ethernet Combo
- CM CAN Combo
- CM LIN Combo
- CM 100 High
- CM 1000 High

Not all features are implemented in all CMs. CM individual restrictions are documented inside the user manuals.

CMs support the official TECMP EtherType (0x99FE). This EtherType is used as reference to transmit the IVN data collected by the CM to an endpoint device connected via Ethernet. The device can be a computer, an Ethernet switch for network connection, or a data logger module. These devices are called "sinks" later in this document.

CMs are used as building blocks in cutting-edge distributed data logging networks made of one or more sinks aggregating the TECMP messages generated inside the measurement network. This logging network is scalable and is based on IT standards.

1.2 Purpose of the Document

This document describes the functionality of the Technica Engineering Capture Module interface. The functions are accessible via the Technically Enhanced Capture Module Protocol (TECMP) published by Technica Engineering and described in this document.

This document defines the data format in which the IVN captured data are transmitted to data sink and how data can be replayed on an IVN. The position of metadata like timestamp and other information related to the source or direction of data are also covered in this document.

1.3 Available Resources

Wireshark provides a native TECMP dissection since v3.4. An overview of the TECMP packet dissection inside Wireshark is provided in the next illustration.

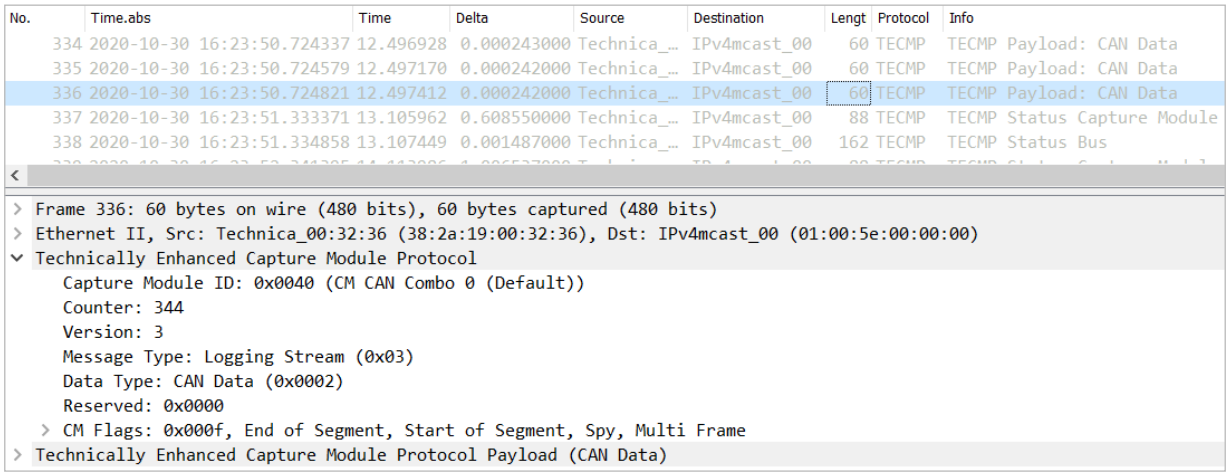


Figure 1 — Example of TECMP packet dissection inside Wireshark (CAN)

1.4 TECMP Function and Scope.

The TECMP protocol is defined by the following key points:

- A specific data format.
- The support of IVN data capturing.
- The support of IVN data replaying
- Specific status and control messages.
- A time synchronization mechanism.

An overview of a system setup comprising multiple CMs and a logger module is represented in Figure 2.

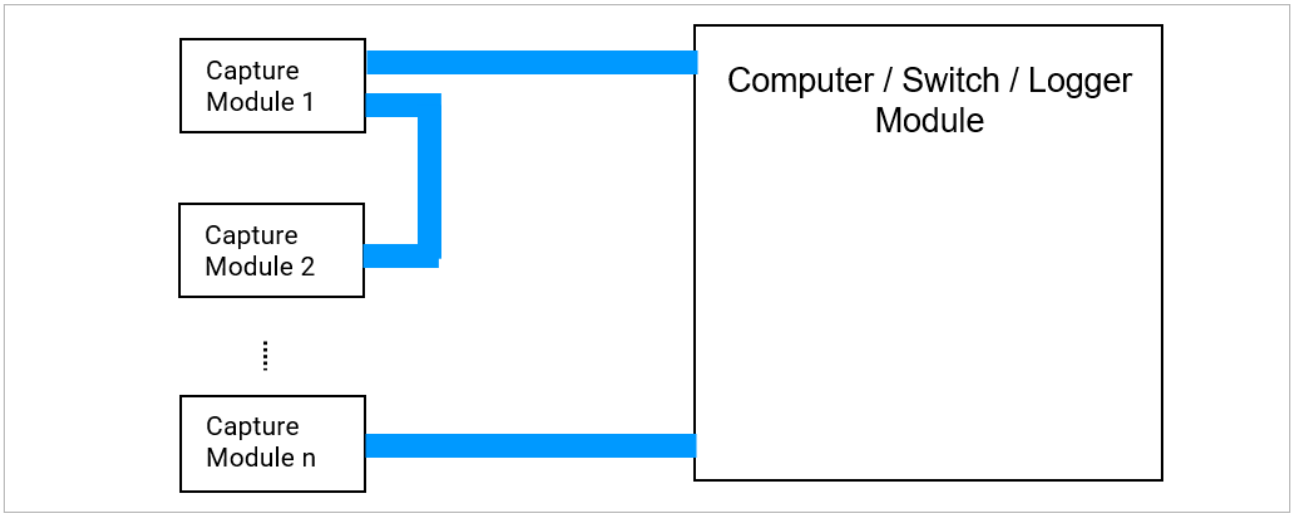


Figure 2 — Overview of a system setup

The following TECMP functionalities are covered by this specification:

- Data transport from the CM to an Ethernet network sink (computer, logger…)
- Data transport to the CM from an Ethernet network sink.
- Remote monitoring and control.
- Time synchronization of the CMs: 802.1AS Automotive Profile (Avnu).

1.5 Glossary

Abbreviation	Definition / Explanation
10BASE-T1S	10 Mbps Baseband Single-Pair Automotive Ethernet Short-link
ACK-DEL	Acknowledgment Delimiter
Avnu	Avnu Alliance https://avnu.org/
BRS	Bit Rate Switch (CAN-FD)
CAN	Controller Area Network
CAN-FD	CAN with Flexible Data-rate
CAS	Collision Avoidance Symbol (FlexRay)
CM	Capture Module
Control Message	A method to trigger events. Events can be triggered by a CM or can be received by a CM. A Control Message is identified by a given Control Message ID (see Section 2.7.4, “Control Message ID”).
CRC	Cyclic redundancy check
CRC-DEL	CRC Delimiter
DDR	Double Data Rate
EoS	End of Segment
Eth	Ethernet
ESDERR	End-of-Stream Delimiter Error
ESDJAB	End-of-Stream Jabber
EOF	End of Frame
ESI	Error State Indicator (CAN-FD)
FCS	Frame Check Sequence
FR	FlexRay
GPIO	General Purpose Input/Output
HW	Hardware
I2C	Inter-Integrated Circuit
ID	Identifier
IDE	Identifier Extension (CAN)
IEEE 802.1AS	IEEE Precision Time Protocol (PTP) Automotive Profile
ILaS	ISELED Light and Sensor network
IVN	In-Vehicle Network
LIN	Local Interconnect Network
LSB	Least Significant Byte(s)
MAC	Media Access Control (Address)
MSB	Most Significant Byte(s)
MII	Media Independent Interface

Abbreviation	Definition / Explanation
PLCA	Physical Layer Coordination Access
PLP	Probe Log Protocol
POC	Protocol Operation Control
PPI	Payload Preamble Indicator (FlexRay)
PRE	Preamble
PTP	Precision Time Protocol
RS232	Recommended Standard 232
RTR	Remote Transmission Request
SFD	Start of Frame Delimiter
SerDes	Serializer/Deserializer
SFP	Small Form-factor Pluggable
SoS	Start of Segment
SPY	Spy Mode
Status Message	A dedicated message sent by the CM in direction of the Ethernet Capture interface (network, computer, logger).
TECMP	Technically Enhanced Capture Module Protocol
Tx	Transmission
UART	Universal Asynchronous Receiver/Transmitter
UINT	Unsigned Integer
VLAN	Virtual Local Area Network (IEEE 802.1Q)
WUP	Wake-Up Pattern/Signal
WUS	Wake-Up Symbol (FlexRay)
Rx	Reception

Table 1 — Glossary

2 Feature Specification

2.1 Foreword

The TECMP protocol enables the recording and replaying of data via a decentralized network based on Ethernet as physical layer. Figure 3 shows an overview of the physical and logical interfaces between the CM and a computer or another Ethernet network components like a datalogger or a switch. For details of the physical interfaces consult the user manual of the specific CM.

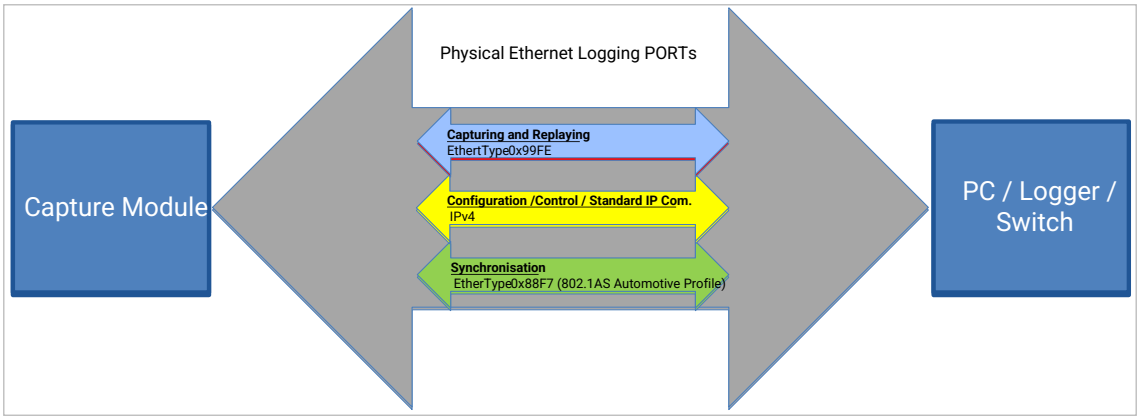


Figure 3 — Logical/physical Interfaces

CMs may provide one or more additional interfaces which can be used to connect CMs with each other in a cascading structure. Based on this feature, the functionality of a CM can be extended with another CM.

2.2 Capturing IVN Data

The main feature of the TECMP protocol is the recording of data of in-vehicle communication systems on a logging network based on Ethernet. Figure 4 presents a simple Logging network made of one CM and one desktop PC:

1. The IVN data (here originating from a CAN, CAN with Flexible Data-rate (CAN-FD) and FlexRay buses) are captured by the CM
2. The captured data is transmitted to the sink in the form of one or more TECMP messages (a.k.a. Logging Stream).

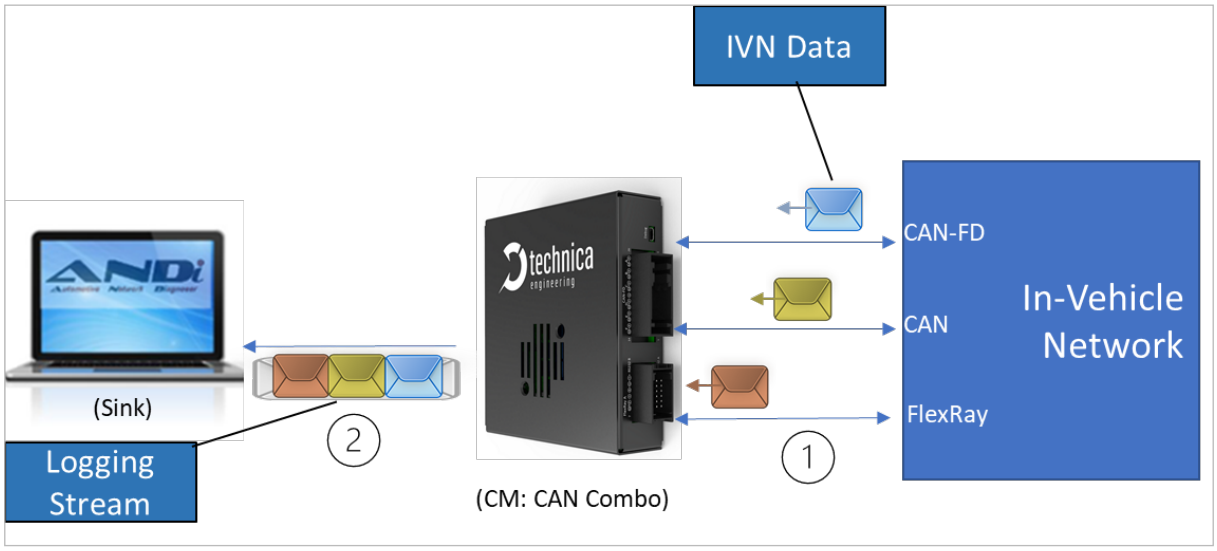


Figure 4 — Capturing IVN Data

For more information on the capturing of data, consult Section 2.4.3, “Capture specification (Message Type = 0x03 “Logging Stream”)”.

2.3 Replaying IVN Data

Additionally, the TECMP protocol supports the transmission of messages into in-vehicle communication systems (Figure 5):

1. The data to transmit on the IVN is received by the CM in the form of one or more TECMP messages (a.k.a. Replay Data).
2. The data to transmit on the IVN is sent out by the CM accordingly to the bus technology (Arbitration on CAN(-FD) and scheduling policy on FlexRay.)

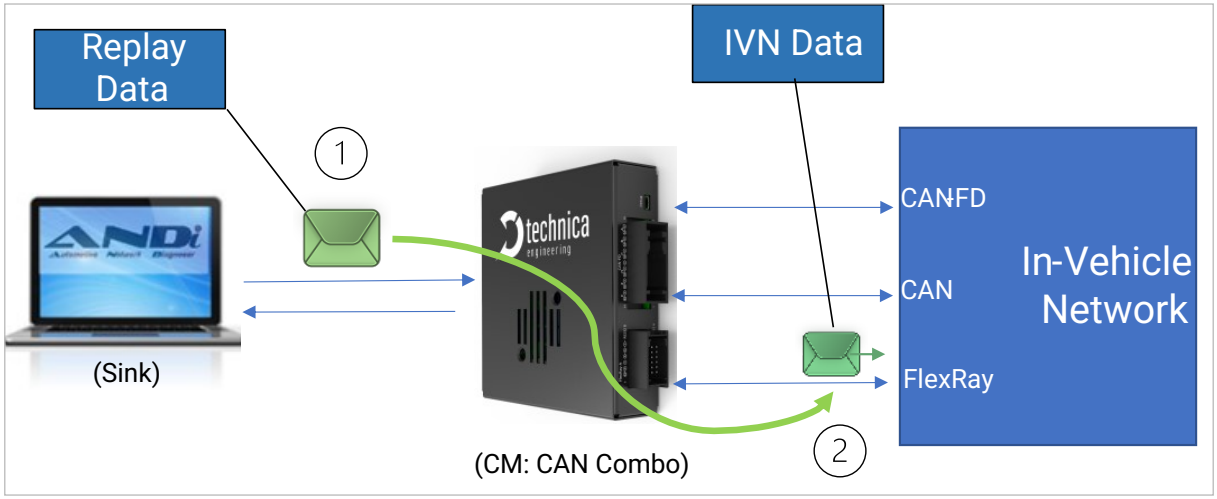


Figure 5 — Replaying data onto the IVN

For more information on the replaying of data, consult Section 2.4.3.10, “TECMP Ethernet RAW (Data Type = 0x0081)”.

2.4 Data Format

A TECMP frame is a standard Ethernet II transport layer frame (IEEE 802.3) with the EtherType 0x99FE and containing a TECMP Header. No additional transport protocol is required.

Data of non-Ethernet communication systems as well as of Ethernet-based systems are transported in Ethernet frames and identified using TECMP. Ethernet frames are transported without change of their payload or header. All messages are left intact. This ensures that analysis tools can easily access the unchanged data, if support of TECMP was added or data was converted.

The data format is defined as follows:

PRE/ SFD	DST MAC	SRC MAC	Outer 802.1Q Header (optional)	Inner 802.1Q Header (optional)	Ether Typ 0x99FE	TECMP Header	Data	FCS
8 Bytes	6 Bytes	6 Bytes	4 Bytes	4 Bytes	2 Bytes	28 Bytes	Up to n Bytes (n <= 1500 Bytes)	4 Bytes

Table 2 — Data Format of a TECMP message

Field Name	Description
PRE / SFD	Preamble and Start Frame Delimiter according to IEEE Standard 802.3.
DST MAC	The destination MAC address is by default assigned to the multicast address: 01: 00: 5E: 00: 00: 00. The value can be updated during configuration.
SRC MAC	The source MAC address carries the MAC address of the CM. The MAC addresses are unique for each delivered hardware.
VLAN 802.1Q	Optional VLAN fields are provided for possible future extensions. • Outer 802.1Q header (optional): Tagging according to IEEE 802.1Q must be supported by all components. The value of the tag must be configurable in the CM. • Inner 802.1Q header (optional): Tagging according to IEEE 802.1Q, must be supported by all components. The value of the tag must be configurable in the CM.
EtherType	0x99FE: TECMP - Identifies this packet as TECMP packet
TECMP header	see Section 2.4.1, “TECMP Header”
Data	see Section 2.4.3, “Capture specification (Message Type = 0x03 “Logging Stream”)”, Section 2.4.3.10, “TECMP Ethernet RAW (Data Type = 0x0081)” and Section 2.4.5, “Packetizing IVN messages”
FCS	CRC checksum, calculated according to IEEE Standard 802.3

Table 3 — TECMP message fields description

2.4.1 TECMP Header

The header structure is identical for all TECMP messages exchanged between CM and sink. Figure 6 shows the position of the header compared to the IVN data position (Data). The header can be separated in two parts:

- A global header section (12 Bytes) from the field Device ID to the field Device Flags.
- A data header section (16 Bytes) from the field Interface ID to the field Data Flags.

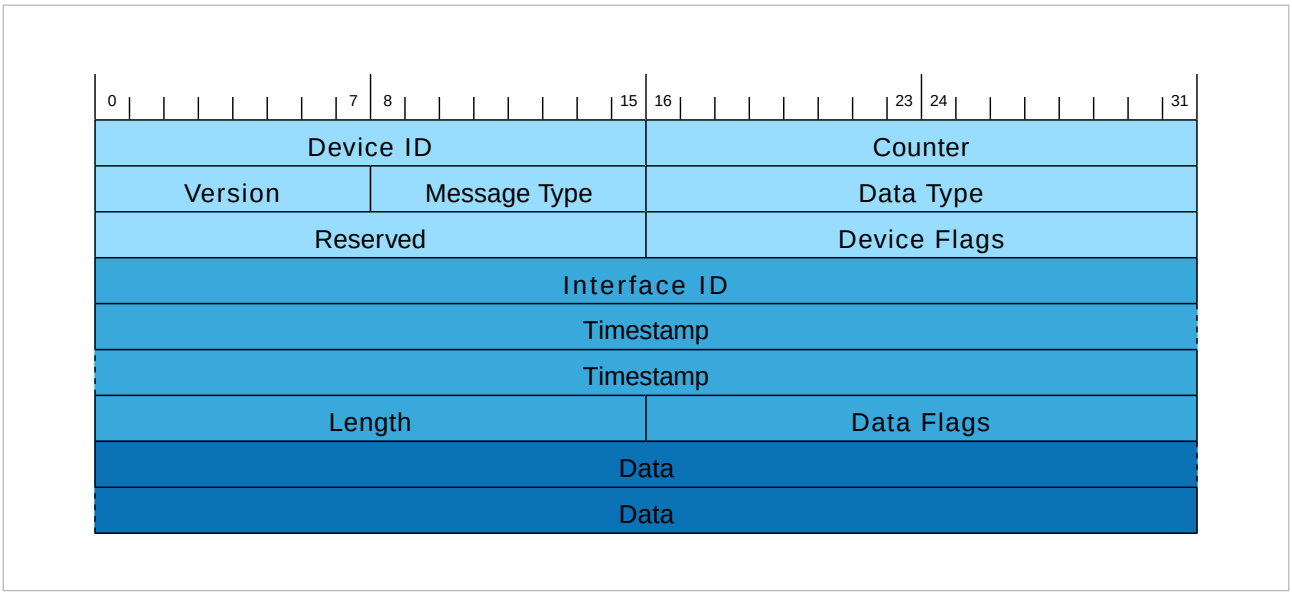


Figure 6 — Overview of a vehicle data frame in a TECMP frame

Figure 7 shows the use of both sections when multiple Data are transmitted into a single TECMP message. In this case, the global header section is used as in Figure 6 while the data header section is repeated prior to every new Data. For more information, refer to Section 2.4.5, “Packetizing IVN messages”.

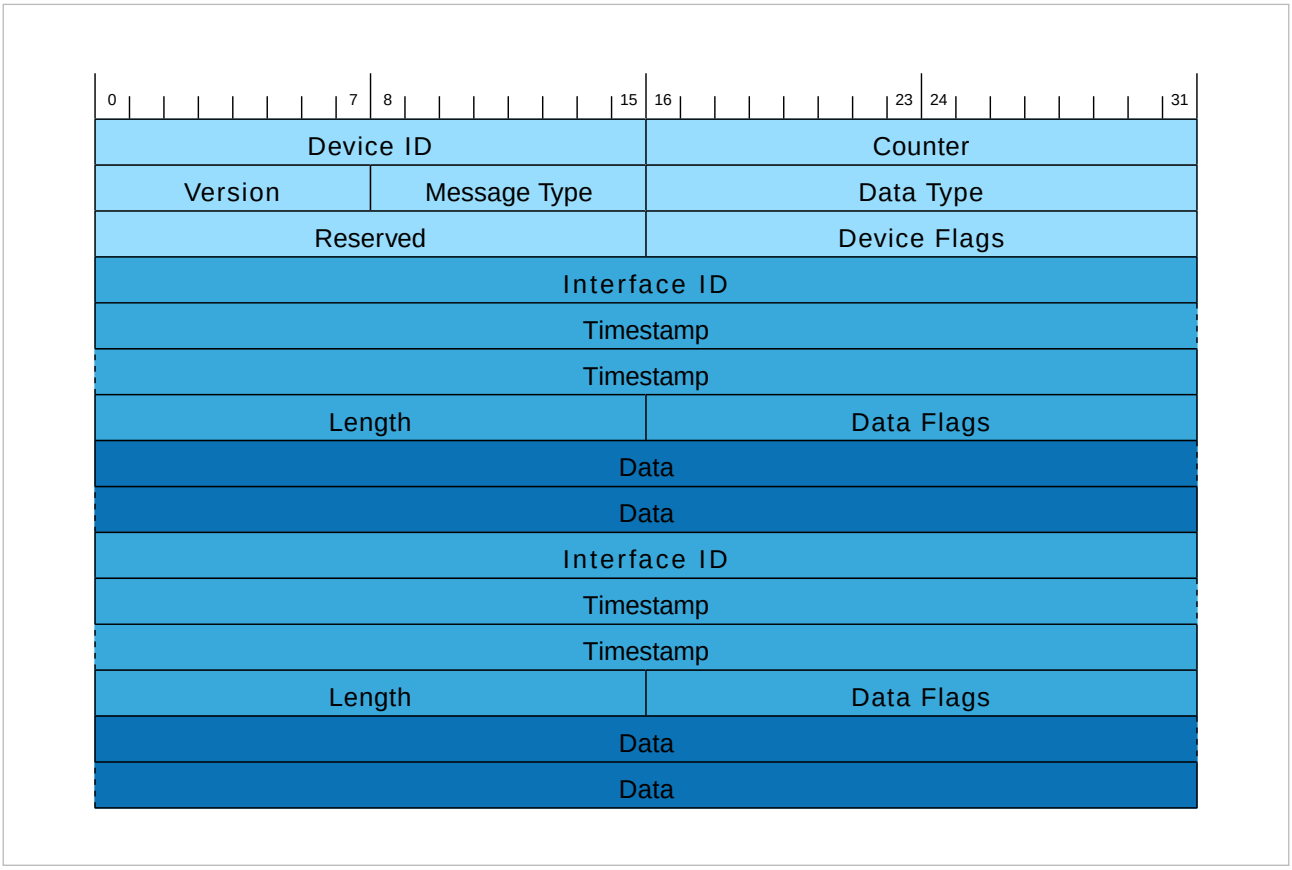


Figure 7 — Overview of multiple vehicle data frames packetized in a TECMP frame

Packing multiple vehicle frames is only done for the same Data Type.

All number types are UINT and encoded in big-endian, if not specified otherwise.

The structure and description of the TECMP Header is presented in Table 4. A distinction between the two cases capturing data/replaying data is made whenever a field of the header has a different description in both cases.

Field Name	Data Type	Description
Device ID	UINT16	The first upper Byte is set to 0x00. The second Byte defines the CM unique ID. This ID can be used to identify different devices if they are used in the same measurement network (see Table 43). Up to 256 different devices can be uniquely identified inside the measurement network. **When capturing data** (Message Type = “Logging Stream”): This field identifies the CM from which the TECMP frame was sent: its source. **When replaying data** (Message Type = “Replay data”): Unused. This field is set to: “0x0000” for every message.
Counter	UINT16	**When capturing data** (Message Type = “Logging Stream”): This field provides a mean to check the non-interruption (loss) of messages sent by the CM. If necessary, the exact number of lost messages can be computed. The counter value is incremented monotonically for each TECMP frame sent from a CM. The counter starts at 0x0001 when the first TECMP frame after power-up is transmitted and increases by 1 at every new transmission. The counter restarts from 0x0000 when reaching the maximum value 0xFFFF. If a CM has several physical interfaces to the sink (Ethernet links), a separate counter is implemented on each interface. The value of this field is only valid for a given Device ID or a given HW interface of the Device ID. **When replaying data** (Message Type = “Replay data”): This field is set to: `0x0000` for every message. Received requests are no counted by the CM.
Version	UINT8	Version of the TECMP protocol used. This value is set to 3.
Message Type	UINT8	Type of TECMP Message: Status, Control, …. See Table 41.
Data Type	UINT16	Description of the Type of Data (e.g., CAN, LIN, FlexRay, Ethernet, Voltage, Current, …). See Table 42.
Reserved	UINT16	Reserved for future extension of the protocol: Set to 0x00 and ignored on receiving.

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>Device Flags</td><td>UINT16</td><td>When capturing data (Message Type = “Logging Stream”):

This field provides additional information on the device.

<table>
<tr><th>Offset</th><th>Name</th><th>Description</th></tr>
<tr><td>0</td><td>EoS</td><td>End of Segment</td></tr>
<tr><td>1</td><td>SoS</td><td>Start of Segment</td></tr>
<tr><td>2</td><td>SPY</td><td>Trigger additional actions on the sink (sink-dependent). Standard value: ‘1’ (none).</td></tr>
<tr><td>3</td><td>Multi-Frame</td><td>’1’: Indicates that the CM can send multiple Interface IDs in a TECMP frame. See Figure 14 in Section 2.4.5, “Packetizing IVN messages”.</td></tr>
<tr><td>14 – 4</td><td>Reserved</td><td>Reserved for future use.</td></tr>
<tr><td>15</td><td>Device overflow</td><td>The CM has lost data during transmission to the Sink.</td></tr>
</table>

When replaying data (Message Type = “Replay data”): Reserved: 0x0000.</td></tr>
<tr><td>Interface ID</td><td>UINT32</td><td>ID that uniquely identifies the log data / bus / link on the IVN.

Status and Control messages have dedicated Interface ID value (Table 46) to ease filtering at this level.</td></tr>
<tr><td>Timestamp</td><td>UINT64</td><td>This field provides the 62-bits timestamp information reference of the TECMP frame, as well as a 2-bits timestamp flag section.

<table>
<tr><th>Bit</th><th>Name</th></tr>
<tr><td>61–0</td><td>Timestamp</td></tr>
<tr><td>62</td><td>Timestamp Recalculation flag</td></tr>
<tr><td>63</td><td>Synchronization flag</td></tr>
</table></td></tr>
<tr><td>Length</td><td>UINT16</td><td>Length of the Data field. The length of the Length and Data Flags fields are not included. Any additional 0-padding Byte added at the end of the TECMP message is not part of the TECMP message itself. It is consequently not considered by the Length field.</td></tr>
<tr><td>Data Flags</td><td>UINT16</td><td>This field provides additional information regarding the recording of the source network when capturing data (Message Type = “Logging Stream”) or regarding the transmission request when replaying data (Message Type = “Replay Data”). The Data Flags refer to the messages transmitted on the IVN (Ethernet, FlexRay, CAN, CAN-FD, LIN…).

See Data type specific definition</td></tr>
</table>

Field Name	Data Type	Description
Data	Variable	• Data transferred from the vehicle IVN **when capturing data** (Message Type = “Logging Stream”). See Section 2.4.3, “Capture specification (Message Type = 0x03 “Logging Stream”)” • Data transferred to the vehicle IVN **when replaying data** (Message Type = “Replay Data”). See Section 2.3, “Replaying IVN Data”

Table 4 — Structure and description of the TECMP Header

2.4.2 Frame overview

Figure 8 represents the overall frame sent towards the sink (between SFD and FCS).

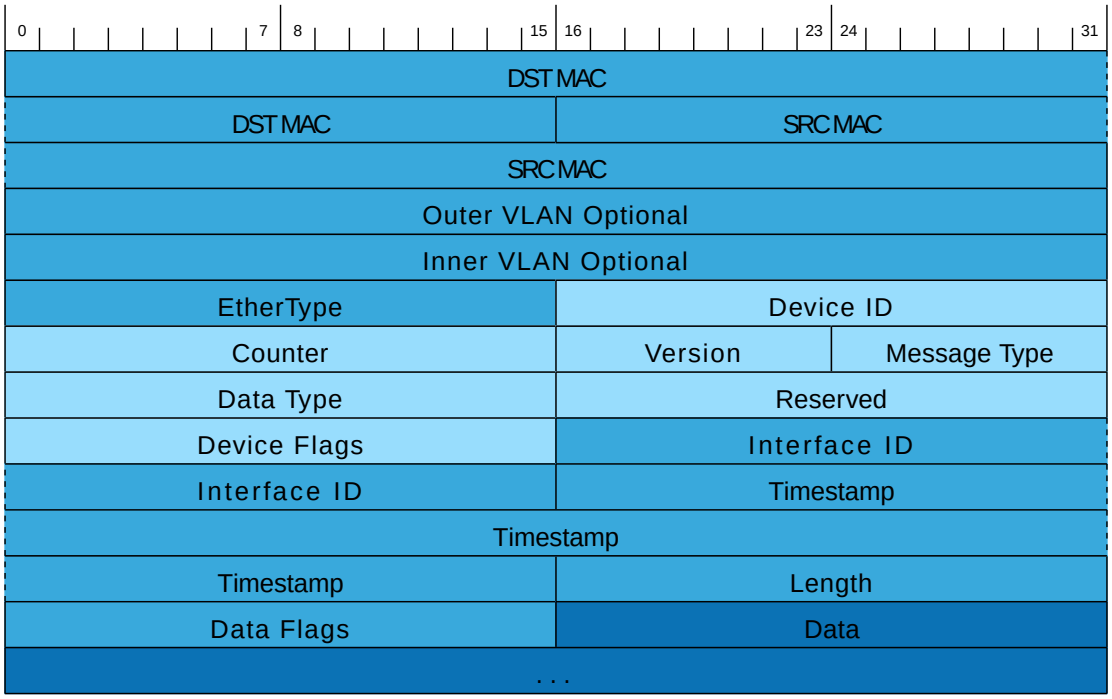


Figure 8 — TECMP frame overview

2.4.3 Capture specification (Message Type = 0x03 “Logging Stream”)

To simplify the data processing and to transmit data as efficiently as possible, data are transferred in the same coding as in the “original” vehicle’s internal buses (IVN). The recorded data are marked with a corresponding timestamp and transferred to the other sinks (e.g. data logger) or to the logging network. The coding details of the data are specified in the next sections.

For efficient transmission, multiple recorded messages can be aggregated into a larger packet

(see Section 2.4.5, “Packetizing IVN messages”). The maximum payload length is limited to 1500 Bytes.

In this section, the Message Type value is 0x3.

In the following, if a special reception condition like an error for which no valid payload can be captured or like a given pattern (WUP for LIN or WUS for FlexRay or instance) then all the field of the Data section are set to 0x0 (for instance the CAN ID for CAN, any payload length…).

2.4.3.1 TECMP CAN Rx (Data Type = 0x0002)

Data Flags

These flags provide information regarding the CAN bus and reception condition.

Bit	Name	Value	Description
0	ACK	**0:** no acknowledgement **1:** acknowledgement	ACK flag.
1	RTR	**0:** data frame **1:** remote frame	Remote Transmission Request flag.
2	IDE	**0:** 11-bit ID **1:** 29-bit ID	Identifier Extension flag.
3	ERR	**0:** no Error Frame received **1:** Error Frame received	Error frame flag.
4	BIT_STUFF error	**0:** no bit stuff error detected **1:** bit stuff error detected	Bit stuffing error flag.
5	CRC_DEL error	**0:** no CRC delimiter error **1:** CRC delimiter error	CRC delimiter error flag.
6	ACK_DEL error	**0:** no acknowledgment delimiter error **1:** acknowledgment delimiter error	ACK delimiter error flag.
7	EOF error	**0:** no end of frame field error **1:** end of frame field error	End of frame error flag.
12 – 8	RESERVED	0	Unused.
13	CRC error	**0:** no CRC error **1:** CRC error detected	CRC error flag.
14	TX	**0:** frame not transmitted by the CM **1:** frame transmitted by the CM	Transmit flag.
15	Overflow	**0:** no message loss detected **1:** one or more messages lost	Overflow flag.

Table 5 — CAN Rx Data Flags

Data

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>CAN ID</td><td>UINT32</td><td>32-bit value containing the CAN identifier together with the Identifier Extension (IDE) bit:

<table><tr><th>Bit(s)</th><th>Name</th><th>Description</th></tr><tr><td>28 – 0</td><td>CAN ID</td><td>CAN Identifier (11 – 29 bits). Unused bits are set to ‘0’.</td></tr><tr><td>30 – 29</td><td>RESERVED</td><td>‘0’: unused.</td></tr><tr><td>31</td><td>IDE</td><td></td></tr></table></td></tr>
<tr><td>Payload Length</td><td>UINT8</td><td>Length of the CAN payload (in Bytes): 0, 1, 2, 3, 4, 5, 6, 7, 8.</td></tr>
<tr><td>Payload</td><td>UINT8 [0-8]</td><td>Payload Bytes (up to 8 Bytes), depending on the Payload Length.</td></tr>
<tr><td>CRC</td><td>UINT8 [2 Bytes]</td><td>CRC (15-bits).

The unused upper bit is filled with ‘0’.</td></tr>
</table>

2.4.3.2 TECMP CAN-FD Rx (Data Type = 0x0003)

Data Flags

These flags provide information regarding the CAN-FD bus and reception condition.

Bit	Name	Value	Description
0	ACK	**0:** no acknowledgement **1:** acknowledgement	ACK flag.
1	ESI	**0:** error node is passive **1:** error node is active	Error State Indicator flag.
2	IDE	**0:** 11 bit ID **1:** 29 bit ID	Identifier Extension flag.
3	ERR	**0:** no Error Frame received **1:** Error Frame received	Error Frame flag. No payload is transported when set; Payload length is set to 0.
4	BRS	**0:** bit rate not switched **1:** bit rate switched	Bit Rate Switch flag. Indicates whether the bit rate was switched from the nominal arbitration phase rate to the data phase rate.
5	BIT_STUFF error	**0:** no bit stuff error detected **1:** bit stuff error detected	Bit Stuff Error flag. Set when an Error Frame is received.
6	CRC_DEL error	**0:** no CRC delimiter error **1:** CRC delimiter error	CRC Delimiter Error flag. Set to 0 when not supported.
7	ACK_DEL error	**0:** no acknowledgment delimiter error **1:** acknowledgment delimiter error	ACK Delimiter Error flag. Set to 0 when not supported.
8	EOF error	**0:** no end of frame field error **1:** end of frame field error	End of Frame Error flag. Set to 0 when not supported.

Bit	Name	Value	Description
9–12	RESERVED	0	Unused.
13	CRC error	**0:** no CRC error found **1:** CRC error found	CRC Error flag.
14	Tx	**0:** frame not transmitted by the CM **1:** frame transmitted by the CM	Transmit flag. Indicates whether the logged frame was transmitted by the CM itself.
15	Overflow	**0:** no message loss detected **1:** one or more messages lost	Overflow flag. Indicates message loss on this interface during recording.

Table 6 — CAN-FD Rx Data Flags

Data

Field Name	Data Type	Description
CAN ID	UINT32	32-bit value containing the received CAN identifier together with the Identifier Extension (IDE) bit: **Bit(s)** / **Name** / **Description** 28 – 0 / CAN ID / CAN Identifier (11 – 29 bits). Unused bits are set to '0'. 30 – 29 / RESERVED / '0': unused. 31 / IDE /
Payload Length	UINT8	Length of the CAN payload (in Bytes): 0, 1, 2, 3, 4, 5, 6, 7, 8, 12, 16, 20, 24, 32, 48, 64.
Payload	UINT8 [0-64]	Bytes of the payload (up to 64 Bytes), depending on the Payload Length.
CRC	UINT8 [3 Bytes]	Received CRC. Note: The 17-bits long CRC is used for FD Frames with a data field up to 16 Bytes while the 21-bits long CRC is used for FD Frames with a data field longer than 16 Bytes. The unused upper bits are filled with '0',

2.4.3.3 TECMP LIN Rx (Data Type = 0x0004)

Data Flags

These flags provide information regarding the LIN bus and reception condition.

Bit	Name	Value	Description
0	Collision error	**0:** no collision **1:** collision detected	Collision on LIN bus.
1	Parity error	**0:** no parity error **1:** parity error detected	Parity error in the Protected ID.
2	No Slave Response	**0:** no issue detected **1:** no slave response	No response from a slave on a request from a master.
7–3	RESERVED	0	Unused.
8	WUP	**0:** no wake-up detection **1:** wake-up signal detected	Detection of a wake-up signal (including a dominant state between 250 µs and 5 ms for LIN 2.0 or exactly 8 dominant bits for LIN 1.3).
9	Short_WUP	**0:** no short wake-up detection **1:** short wake-up signal detected	Detection of a short wake-up signal (including a dominant state between 150 µs and 249 µs for LIN 2.0). This is always 0 for LIN 1.3.
10	SLEEP	**0:** no go-to-sleep command detected **1:** go-to-sleep command detected	Detection of a go-to-sleep command.
12–11	RESERVED	0	Unused.
13	Checksum error	**0:** no checksum error **1:** checksum error detected	Checksum error found. The frame may be corrupted or the wrong checksum model was used.
14	Tx	**0:** frame not transmitted by CM **1:** frame transmitted by CM	The frame logged on the interface was transmitted by the CM itself.
15	Overflow	**0:** no message loss **1:** message loss detected	One or more messages on this interface were lost during recording.

Table 7 — LIN Rx Data Flags

Data

Field Name	Data Type	Description
LIN ID	UINT8	Frame identifier: • Bit 7 – Bit 6 : Reserved (0x0) • Bit 5 – Bit 0 : LIN ID.
Payload Length	UINT8	LIN Length of the LIN payload (1,2,4 or 8 Bytes).
Payload	UINT8 [0-8]	Payload Bytes (up to 8 Bytes), depending on the Payload Length.
Checksum	UINT8	LIN checksum (For frames with a ‘1’ value in the “No Slave Response” DataFlag, this field does not exist.)

2.4.3.4 TECMP FlexRay Rx (Data Type = 0x0008)

Data Flags

These flags provide information regarding the FlexRay bus and reception condition.

Bit	Name	Value	Description
0	NF	**0:** not a null frame **1:** null frame	Null frame indicator.
1	SF	**0:** not a startup frame **1:** startup frame	Startup frame indicator.
2	Sync	**0:** not a sync frame **1:** sync frame	Sync frame indicator.
3	WUS	**0:** not a wakeup symbol **1:** wakeup symbol (no payload)	Wakeup symbol indicator.
4	PPI	**0:** PPI bit not set **1:** PPI bit set	Payload preamble indicator.
5	CAS	**0:** not a collision avoidance symbol **1:** collision avoidance symbol	Collision avoidance symbol indicator.
6–10	RESERVED	0	Unused.
11	Channel	**0:** FlexRay Channel A **1:** FlexRay Channel B	FlexRay channel indicator.
12	Header CRC error	**0:** no header CRC error **1:** header CRC error detected	Header CRC error indicator.
13	Frame CRC error	**0:** no frame CRC error **1:** frame CRC error detected	Frame CRC error indicator.
14	Tx	**0:** not transmitted by the CM **1:** transmitted by the CM	Transmit indicator.
15	Overflow	**0:** no message loss **1:** one or more messages lost	Overflow indicator.

Table 8 — FlexRay Rx Data Flags

Data

Field Name	Data Type	Description
Cycle	UINT8	Cycle in which the frame was received.
Frame ID	UINT16	Frame ID.
Payload Length	UINT8	Length of FlexRay Frame Payload for a specific ID.
Payload	UINT8 [0-254]	Bytes of the payload (up to 254 Bytes), depending on the Payload Length.
Header CRC	UINT16	Header CRC.
Frame CRC	UINT8 [3 Bytes]	Frame CRC.

2.4.3.5 TECMP GPIO (Data Type = 0x000A)

GPIOs are hardware-based pins attached to SerDes devices or similar.

Data Flags

Bit	Name	Value	Description
13–0	RESERVED	0	Unused.
14	Tx	**0:** not transmitted by CM **1:** transmitted by CM	Tx flag.
15	Overflow	**0:** no overflow **1:** overflow	Message loss indicator.

Table 9 — GPIO Data Flags

Data

Data Type	Field Name	Description
UINT8	GPIOs 0 – 7	Captured state of the first group of 8 GPIOs: 0 = Low state 1 = High state
UINT8	GPIOs 8 – 15	Captured state of the second group of 8 GPIOs: 0 = Low state 1 = High state
…		

The number of GPIOs is implementation specific and different devices might have a different number of supported GPIOs.

2.4.3.6 TECMP ILaS Data (Data Type = 0x000E)

Data Flags

Bit	Name	Value	Description
0	CRC Received	**0:** CRC field not present in received message **1:** CRC field present in received message	CRC field presence indicator.
1	Direction	**0:** Downstream (command message) direction **1:** Upstream (response message) direction	Logged frame direction indicator.
2–12	RESERVED	0	Unused.
13	CRC Error	**0:** No CRC Error detected **1:** CRC Error detected	CRC error indicator.
14 – 15	RESERVED	0	Unused.

Table 10 — ILaS Data Flags

Data

Field Name	Data Type	Description
Decoded API command	UINT8	High level command to encode ILaS commands
Decoded Address	UINT16	Captured and Decoded ILaS Address
Decoded Data	UINT8[3]	Captured and Decoded ILaS Data
RAW SDU	UINT8[7]	Captured RAW SDU field from ILaS frames
RAW CRC	UINT16	Captured RAW CRC field from ILaS frames. 0x00 if CRC Enable in Flags is ‘0’

2.4.3.7 TECMP UART / RS232 Rx (Data Type = 0x0010)

Data Flags

These flags detail the UART / RS232 bus and reception condition.

Bit	Name	Value	Description
0	Parity error	**0:** no parity error **1:** parity error	Parity error flag.
1–3	DL	**010:** RS232 with 7 bits characters (logged as 8 bits) **011:** RS232 with 8 bits characters (logged as 8 bits)	Symbol length on RS232.
4–13	RESERVED	0	Unused.
14	Tx	**0:** frame not transmitted by the CM itself **1:** frame transmitted by the CM itself	Transmission direction flag.
15	RESERVED	0	Unused.

Table 11 — UART / RS232 Rx Data Flags

Data

Several data symbols of an RS232 interface are collected inside the Data. In this case, only the user data (7 – 8-bit length per data symbol) is transmitted in one Byte (possibly filled with '0 bits').

A maximum of 1000 RS232 data symbols may be transmitted in one TECMP frame

The timestamping takes place at the first symbol reception on the RS232 line. Configuration of the interface (bitrate, symbol length, parity bit, …) takes place in the CM. Settings relevant to the recording are stored in the Data Flags.

Field Name	Data Type	Description
Payload	UINT8	RS-232 message (1 Byte per RS-232 symbol)
…	..	…
Payload n	UINT8	

2.4.3.8 TECMP Analog Rx (Data Type = 0x0020)

Data Flags

These flags provide information regarding the Analog bus.

Bit	Name	Value	Description
1–0	RESERVED	0	Unused.
4–2	Unit	**000:** Volt (V) **001:** Ampere (A) **010:** Watt (W) **011:** Ampere hour (Ah) **100:** Degree Celsius (°C)	Measurement unit.
6–5	RESERVED	0	Unused.
8–7	Factor	**00:** 0,1 **01:** 0,01 **10:** 0,001 **11:** 0,0001	Scaling factor for measured values.
10–9	RESERVED	0	Unused.
14–11	Sample Time	**0000:** Reserved **0001:** 2500ms **0010:** 1000ms **0011:** 500ms **0100:** 250ms **0101:** 100ms **0110:** 50ms **0111:** 25ms **1000:** 10ms **1001:** 5ms **1010:** 2,5ms **1011:** 1ms **1100:** 0,5ms (optional) **1101:** 0,25ms (optional) **1110:** 0,1ms (optional) **1111:** 0,05ms (optional)	Analog sampling interval.
15	RESERVED	0	Unused.

Table 12 — Analog Rx Data Flags

Data

A maximum of 700 measured values may be transmitted in one TECMP frame

The timestamping takes place when the first analog value is acquired. Configuration of the interface (voltage, current, sample time, …) takes place in the sample. Settings relevant to the recording are stored in the Data Flags.

Field Name	Data Type	Description
Sample 0	INT16	Contains the first value of the recording.
…	INT16	…
Sample N	INT16	Contains the N-th value of the recording.

2.4.3.9 TECMP Ethernet Rx (Data Type = 0x0080)

Data Flags

These flags provide information regarding the reception condition on the Ethernet interface.

Bit	Name	Value	Description
12–0	RESERVED	0	Unused.
13	CRC error	**0:** no CRC error detected **1:** CRC error detected	CRC error flag.
14	TX	**0:** frame was not transmitted by the CM itself **1:** frame was transmitted by the CM itself	Transmit flag.
15	Overflow	**0:** no message loss detected **1:** one or more messages lost during recording	Overflow flag.

Table 13 — Ethernet Rx Data Flags

Data

Ethernet packets of the size of standard frames are completely transferred to the Ethernet payload of the newly generated packet, from MAC Destination up to FCS (Table 14 and Table 15).

Example: A recorded Ethernet Message on an IVN Link.

“data” comes from a transmission of an Automotive Ethernet Packet (Ethernet II):

The payload is a standard Ethernet frame.

Data						
DST MAC	SRC MAC	Outer 802.1Q Header (optional)	Inner 802.1Q Header (optional)	Ether Type	Payload	FCS
6 Bytes	6 Bytes	4 Bytes	4 Bytes	2 Bytes	x Bytes	4 Bytes

Table 14 — Structure of an Ethernet Packet (Ethernet II)

In this case the EtherType of the Ethernet Header is the entry of the message to be recorded from the source network.

Field Name	Data Type	Description
DST MAC	UINT8[6]	Destination MAC address.
SRC MAC	UINT8[6]	Source MAC address.
Outer VLAN Tag (optional)	UINT32	Outer 802.1Q Header.
Inner VLAN Tag (optional)	UINT32	Inner 802.1Q Header.
EtherType	UINT16	Ethernet protocol Type.
Payload	UINT8[x]	Payload Bytes.
FCS	UINT32	Frame Check Sequence.

Table 15 — Description of the fields of an Ethernet Packet (Ethernet II)

2.4.3.10 TECMP Ethernet RAW (Data Type = 0x0081)

Data Flags

These flags provide information regarding the reception condition on the Ethernet interface.

Bit	Name	Value	Description
12–0	RESERVED	0	Unused.
13	CRC error	**0:** no CRC error detected **1:** CRC error detected	CRC error flag.
14	TX	**0:** frame not transmitted by the CM itself **1:** frame transmitted by the CM itself	Transmit direction flag.
15	Overflow	**0:** no message loss detected **1:** one or more messages lost	Overflow flag.

Table 16 — Ethernet RAW Data Flags

Data

The payload is a standard Ethernet frame with preamble and SFD before:

Data								
Preamble	SFD	DST MAC	SRC MAC	Outer 802.1Q Header (optional)	Inner 802.1Q Header (optional)	Ether Type	Payload	FCS
0..7 Bytes	1 Byte	6 Bytes	6 Bytes	4 Bytes	4 Bytes	2 Bytes	x Bytes	4 Bytes

Table 17 — Structure of Ethernet Raw Data

2.4.3.11 TECMP 10BASE-T1S Data (Data Type = 0x0082)

Data Flags

Bit	Name	Value	Description
11–0	RESERVED	0	Unused.
12	PHY Event/Error	**0:** no event/error **1:** event/error reported	PHY event or error indicator.
13	CRC Error	**0:** no CRC error detected **1:** CRC error detected	CRC error indicator.
14	TX	**0:** frame not transmitted by the CM **1:** frame transmitted by the CM	Indicates the frame was transmitted by the capture module itself.
15	Overflow	**0:** no message loss detected **1:** one or more messages lost	Indicates message loss on this interface during recording.

Table 18 — 10BASE-T1S Data Flags

Data

The Data field is a standard Ethernet frame, preceded by a Beacon Timestamp (8 Bytes) field.

Ethernet packets of the size of standard frames are completely transferred to the Ethernet payload of the newly generated packet, from MAC Destination up to FCS (Table 14 and Table 15).

Example: A recorded Ethernet Message on an IVN Link.

Data							
Beacon Timestamp	DST MAC	SRC MAC	Outer 802.1Q Header (optional)	Inner 802.1Q Header (optional)	EtherType	Payload	FCS
8 Bytes	6 Bytes	6 Bytes	4 Bytes	4 Bytes	2 Bytes	x Bytes	4 Bytes

Table 19 — Structure of an Ethernet 10BASE-T1S Packet

In this case, the EtherType of the Ethernet Header is the entry of the message to be recorded from the source network.

Field Name	Data Type	Description
Beacon Timestamp	UINT64	Timestamp of the beacon of the current cycle.
DST MAC	UINT8[6]	Destination MAC address.
SRC MAC	UINT8[6]	Source MAC address.
Outer VLAN Tag (optional)	UINT32	Outer 802.1Q Header.
Inner VLAN Tag (optional)	UINT32	Inner 802.1Q Header.
EtherType	UINT16	Ethernet protocol Type.
Payload	UINT8[x]	Payload Bytes.
Checksum	UINT32	Frame Check Sequence.

Table 20 — Description of the fields of an Ethernet 10BASE-T1S Packet

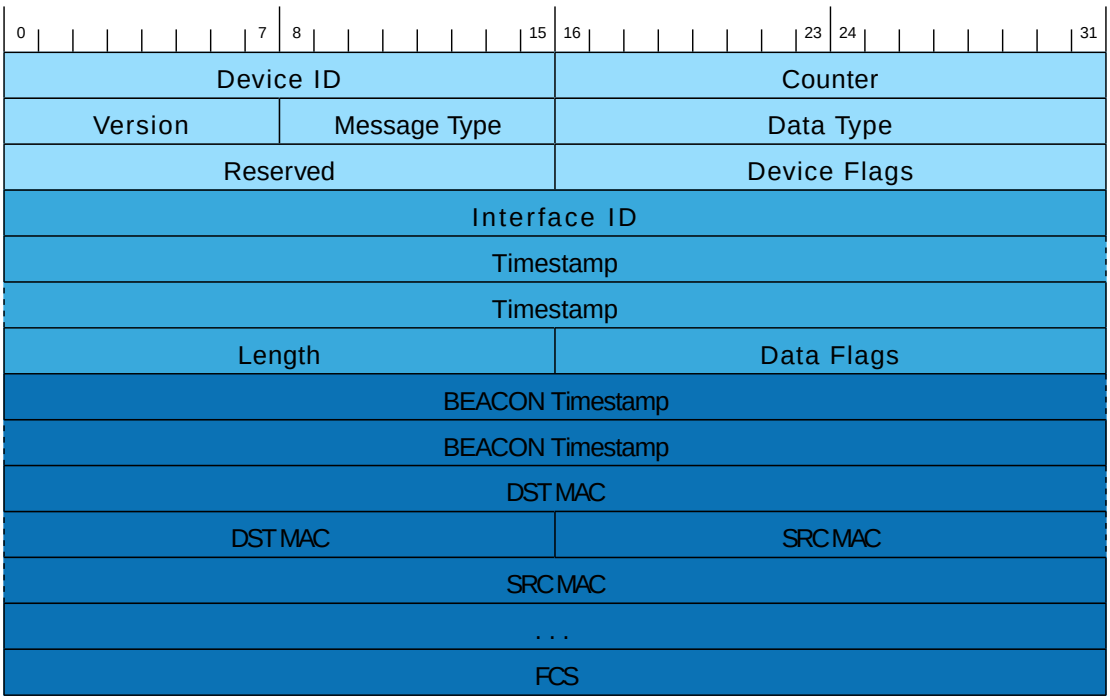


Figure 9 — Overview of a 10BASE-T1S data frame in a TECMP frame



Figure 10 — Overview of multiple 10BASE-T1S data frames packetized in a TECMP frame

2.4.3.12 ENHANCED ETHERNET 10BASE-T1S - PLCA Symbol Logging (Data Type = 0x0083)

The **Ethernet 10BASE-T1S PLCA Symbol Logging** data type (0x0083) is used by TECMP to show a snapshot of the physical-layer symbols.

Timestamp

A 64-bit unsigned integer that records the **exact time** when the first symbol of the logged sequence was detected on the physical line.

Payload

The payload contains the **captured PLCA symbols** in the order they appeared on the wire. Each symbol is originally a 5-bit value on the physical layer. For transport convenience it is stored in a 1-byte value where the 5 significant bits are **7-3** and bits **2-0** are zero-padded.

The symbols are grouped according to the packet-type definitions in the 10BASE-T1S specification. Table 21 shows the packet groups; Table 22 lists the exact 5-bit codes and byte representation as specified in IEEE Std 802.3-2022 clause 147.

Symbols Pattern	Description	Relative packet position
5xBEACON	Coordinator or node without transmission	PRE
2xSYNC + 2xSSD_OK	Node with transmission	PRE
1xESD + 1xESD_OK	Post data symbols	POST
Unknown Symbols	No valid transmission. DECODE_ERR is set to '1'	-

Table 21 — PLCA symbols packet groups

Symbol name	5-bit pattern	1-byte pattern
SYNC/COMMIT	`11000`	`1100 0000 (0xC0)`
ESDERR	`10001`	`1000 1000 (0x88)`
ESD/HB	`01101`	`0110 1000 (0x68)`
ESDOK/ESDBRS	`00111`	`0011 1000 (0x38)`
SSD	`00100`	`0010 0000 (0x20)`
BEACON	`01000`	`0100 0000 (0x40)`

Table 22 — PLCA symbols

Data Flags

A 16-bit field that provides additional status information about the captured symbols.

Bit	Name	Value	Description
14–0	RESERVED	0	Unused.
15	DECODE_ERR	**0:** no error **1:** symbol errors detected	Decode error flag.

Table 23 — 10BASE-T1S PLCA Data Flags

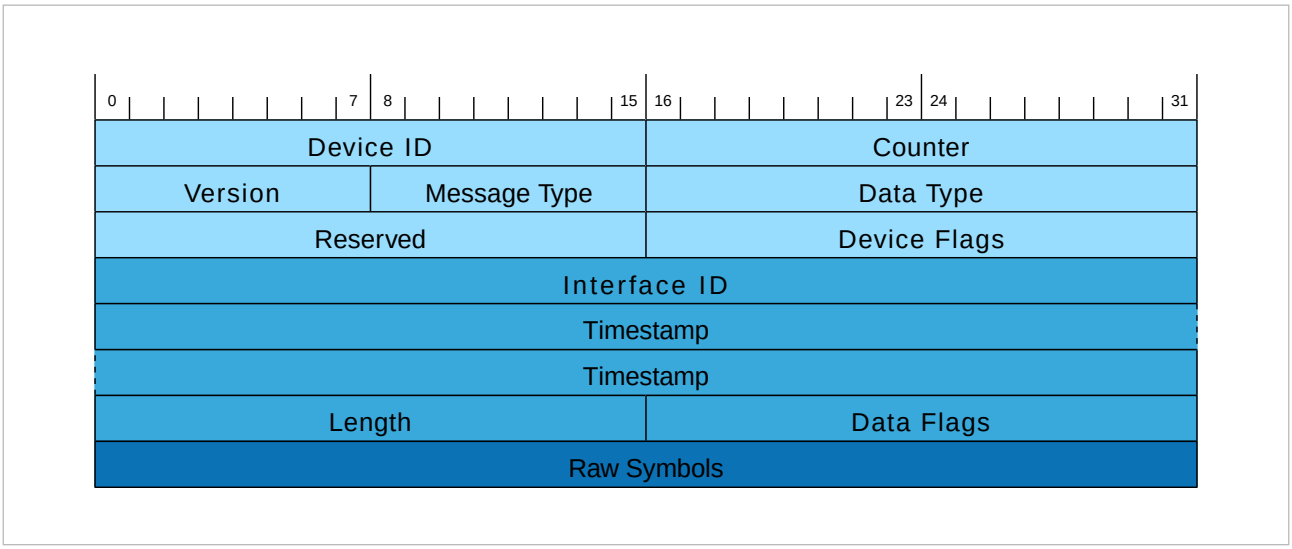


Figure 11 — Enhanced 10BASE-T1S TECMP frame layout

2.4.3.13 TECMP I2C (Data Type = 0x0104)

This data format can transport 7-bit as well as 10-bit addresses in the same way that I2C transports 7-bit and 10-bit addresses.

Data Flags

Bit	Name	Value	Description
0	No ACK error	**0:** no error **1:** no ACK received	Missing ACK error flag.
1–12	RESERVED	0	Unused.
13	RESERVED	0	Unused.
14	Tx	**0:** the frame logged on the interface was not transmitted by the CM itself **1:** the frame logged on the interface was transmitted by the CM itself	Transmission direction flag.
15	Overflow	**0:** no message loss was detected on this interface during recording **1:** one or more messages on this interface were lost during recording	Message loss indicator flag.

Table 24 — I2C Data Flags

Data

The I2C data alternates between address/data byte and control byte.

The values for the control bytes are defined as follows:

- 0x00 NACK
- 0x01 ACK
- 0x02 ACK + STOP
- 0x03 ACK + repeated START

- 0x04 NACK + STOP
- 0x05 NACK + repeated START
- 0x06 – 0xff Reserved

Example:

If on I2C an ACK and then a repeated START (Sr) is seen, the CM will generate a single control byte 0x03 combining both.

2.4.4 Replay specification (Message Type = 0x0A “Replay Data”)

As introduced in Section 2.3, “Replaying IVN Data”, the TECMP protocol supports the replaying of data into an in-vehicle network. A Replay Data example is represented in the next Figure.

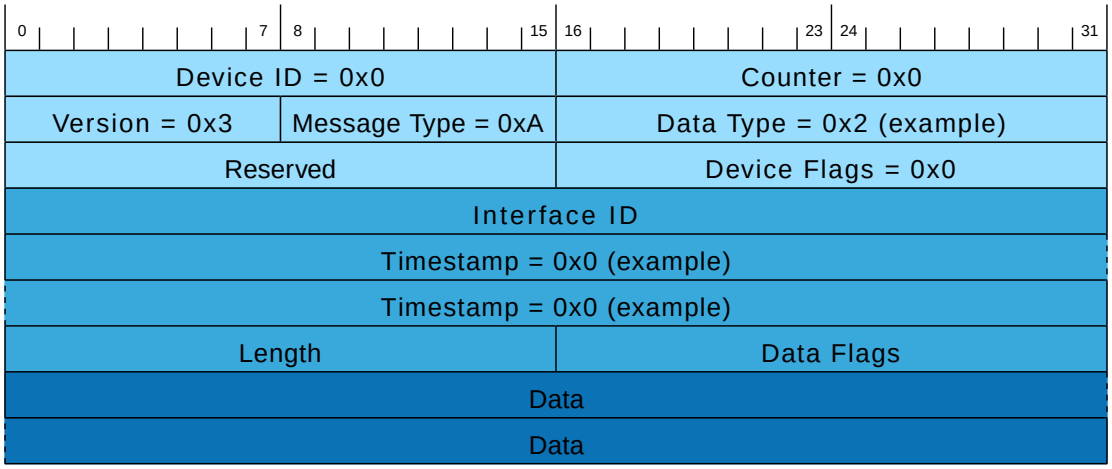


Figure 12 — Replay Data example

In this example, a CAN frame (Data Type = 0x2) shall be replayed (Message Type = 0xA) given the current protocol Version (0x3). The frame shall be sent as soon as possible (Timestamp = 0x0). The supported values of the Data Flags and Data for every Data Type are defined in the next subsections.

For replaying or sending data the Message Type value is 0xA.

2.4.4.1 TECMP CAN Tx (Data Type = 0x0002)

Data Flags

These flags control the transmission properties on the CAN bus.

Bit	Name	Value	Description
0	RESERVED	0x0	Unused.
1	RTR	**0:** data frame **1:** remote frame	Remote frame indicator.
3–2	RESERVED	0x0	Unused.
4	BIT_STUFF error	**0:** bit-stuffing will not be interrupted **1:** a bit stuff error will be generated	Bit stuff error injection.
5	CRC_DEL error	**0:** no CRC delimiter error will be injected **1:** a CRC delimiter error will be injected	CRC delimiter error injection.
6	ACK_DEL error	**0:** no acknowledgment delimiter error will be injected **1:** an acknowledgment delimiter error will be injected	Acknowledgment delimiter error injection.
7	EOF error	**0:** no end of frame field error will be injected **1:** an end of frame field error will be injected	End of frame error injection.
12 – 8	RESERVED	0x0	Unused.
13	Use CRC	**0:** the device will compute the correct value for the transmission **1:** the provided Checksum value will be used for the transmission	CRC source selector.
15 – 14	RESERVED	0x0	Unused.

Table 25 — CAN Tx Data Flags

Data

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>CAN ID (inclusive IDE bit)</td><td>UINT32</td><td>32-bit value containing the CAN identifier together with the Identifier Extension (IDE) bit:
<table>
<tr><th>Bit(s)</th><th>Name</th><th>Description</th></tr>
<tr><td>28 - 0</td><td>CAN ID</td><td>CAN Identifier (11 – 29 bits). Unused bits are set to ‘0’.</td></tr>
<tr><td>30 - 29</td><td>RESERVED</td><td>‘0’: unused</td></tr>
<tr><td>31</td><td>IDE</td><td></td></tr>
</table></td></tr>
<tr><td>Payload Length</td><td>UINT8</td><td>Length of the CAN payload (in Bytes): 0, 1, 2, 3, 4, 5, 6, 7, 8.</td></tr>
<tr><td>Payload</td><td>UINT8 [0-8]</td><td>Payload Bytes (up to 8 Bytes), depending on the Payload Length.</td></tr>
<tr><td>CRC</td><td>UINT8 [2 Bytes]</td><td>Depending on the value of the Use CRC bit (13) inside the Data Flags:
‘0’ : The provided CRC value will not be used. The device will compute the correct value for the transmission.
‘1’ : The provided CRC (15-bits) value will be used for the transmission. The unused upper bit shall be filled with ‘0’: the range 0x8000-0xFFFF is undefined.</td></tr>
</table>

2.4.4.2 TECMP CAN-FD Tx (Data Type = 0x0003)

Data Flags

These flags control the transmission properties on the CAN-FD bus.

Bit	Name	Value	Description
3–0	RESERVED	0x0	Unused.
4	BRS	**0:** bit rate not switched **1:** bit rate switched	Bit Rate Switch flag.
5	BIT_STUFF_error	**0:** bit-stuffing not interrupted **1:** bit stuff error generated	Bit stuff error injection flag.
6	CRC_DEL error	**0:** no CRC delimiter error injected **1:** CRC delimiter error injected	CRC delimiter error injection flag.
7	ACK_DEL error	**0:** no acknowledgment delimiter error injected **1:** acknowledgment delimiter error injected	ACK delimiter error injection flag.
8	EOF error	**0:** no end of frame field error injected **1:** end of frame field error injected	End of frame error injection flag.
12–9	RESERVED	0x0	Unused.
13	Use CRC	**0:** CRC computed by device **1:** provided CRC value used	Use provided CRC value for transmission.
15–14	RESERVED	0x0	Unused.

Table 26 — CAN-FD Tx Data Flags

Data

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>CAN ID (inclusive IDE bit)</td><td>UINT32</td><td>32-bit value containing the received CAN identifier together with the Identifier Extension (IDE) bit:

<table>
<tr><th>Bit(s)</th><th>Name</th><th>Description</th></tr>
<tr><td>28 - 0</td><td>CAN ID</td><td>CAN Identifier (11 – 29 bits). Unused bits are set to ‘0’.</td></tr>
<tr><td>30 - 29</td><td>RESERVED</td><td>‘0’: unused</td></tr>
<tr><td>31</td><td>IDE</td><td></td></tr>
</table>
</td></tr>
<tr><td>Payload Length</td><td>UINT8</td><td>Length of the CAN payload (in Bytes): 0, 1, 2, 3, 4, 5, 6, 7, 8, 12, 16, 20, 24, 32, 48, 64.</td></tr>
<tr><td>Payload</td><td>UINT8 [0-64]</td><td>Bytes of the payload (up to 64 Bytes), depending on the Payload Length.</td></tr>
<tr><td>CRC</td><td>UINT8 [3 Bytes]</td><td>Depending on the value of the Use CRC bit (13) inside the Data Flags:

• ‘0’ : The provided CRC value will not be used. The device will compute the correct value for the transmission.
• ‘1’ : The provided CRC (17/21-bits) value will be used for the transmission. The unused upper bit shall be filled with ‘0’. The 17-bits long CRC is used for FD Frames with a data field up to 16 Bytes long while the 21-bits long CRC is used for FD Frames with a data field longer than 16 Bytes.</td></tr>
</table>

2.4.4.3 TECMP LIN Tx (Data Type = 0x0004)

Data Flags

These flags control the transmission properties on the LIN bus.

Bit	Name	Value	Description
0	RESERVED	0x0	Unused.
1	Use_Parity	**0:** parity bits computed by the device **1:** provided parity bits used for transmission	Use parity flag.
2–6	RESERVED	0x0	Unused.
7	RST_LIN_TABLE	**0:** no operation **1:** clear all saved LIN responses to a given master ID request; all other data flags and data values will be ignored	Reset LIN table (applicable only if LIN interface is configured as LIN slave).
8	WUP	**0:** no wake up transmission **1:** transmit a wake up signal (dominant state between 250 µs and 5 ms for LIN 2.0, or exactly 8 dominant bits for LIN 1.3)	Wake up pulse flag.
9	Short_WUP	**0:** no short wake up transmission **1:** transmit a short wake up signal (dominant state between 150 µs and 249 µs); only supported for LIN 2.0	Short wake up pulse flag.
10 – 12	RESERVED	0x0	Unused.
13	Use_Checksum	**0:** checksum computed by the device **1:** provided checksum used for transmission	Use checksum flag.
14 – 15	RESERVED	0x0	Unused.

Table 27 — LIN Tx Data Flags

Data

Field Name	Data Type	Description
LIN ID	UINT8	Frame identifier: • Bit 7 – Bit 6: Depending on the value of the “Use Parity” bit (1) inside the Data Flags: • ‘0’: The Parity bits will be computed by the device. • ‘1’: The provided Parity bits will be used for the transmission. ○ Bit 5 – Bit 0: LIN ID •
Payload Length	UINT8	LIN Length of the LIN payload (1,2,4 or 8 Bytes). A 0-length value can be used to remove the slave response configuration entry associated with a given LIN ID, from the CM memory.
Payload	UINT8 [0-8]	Payload Bytes (up to 8 Bytes), depending on the Payload Length.
Checksum	UINT8	Depending on the value of the Use Checksum bit (13) inside the Data Flags: • ‘0’: The Checksum will be computed by the device. • ‘1’: The provided Checksum value (0x00-0xFF) will be used for the transmission.

2.4.4.4 TECMP FlexRay Tx (Data Type = 0x0008)

Data Flags

These flags control the transmission properties on the FlexRay bus.

Bit	Name	Value	Description
6–0	RESERVED	0x0	Unused.
9–7	Tx Mode	See below	Transmission mode. **0x0:** Reserved. **0x1:** Single Shot — the configured frame is transmitted exactly once. **0x2:** Continuous — the configured frame remains valid until explicitly marked invalid (TX-NONE). **0x3:** TX-NONE — the configured frame is explicitly marked as invalid. **0x4–0x7:** Undefined.
10	RESERVED	0x0	Unused.
11	Channel	See below	FlexRay channel selection. **0:** FlexRay Channel A. **1:** FlexRay Channel B.
12	Use Header CRC	See below	Controls whether the provided Header CRC is used for transmission. **0:** The provided Checksum value will not be used; the device computes the correct value. **1:** The provided Checksum value (0x0000–0x07FF) will be used; any other value (0x0800–0x0FFF) causes the device to compute the correct value instead.
15–13	RESERVED	0x0	Unused.

Table 28 — FlexRay Tx Data Flags

Data

Field Name	Data Type	Description
Cycle	UINT8	Cycle Offset
Frame ID	UINT16	Frame ID
Payload Length	UINT8	Length of FlexRay Frame Payload for a specific ID
Payload	UINT8 [0–254]	Bytes of the payload (up to 254 Bytes), depending on the Payload Length.
Cycle Repetition	UINT8	Valid Values: 1, 2, 4, 8, 16, 32, 64.
Header CRC	UINT16	Depending on the value of the Use Header CRC bit (12) inside the Data Flags: • '0' ○ The provided Header CRC value will not be used. The device will compute the correct value for the transmission. • '1' ○ 0x0000-0x07FF: The provided Header CRC value will be used for the transmission. ○ 0x0800-0xFFFF: Undefined: The device will compute the correct value for the transmission.



The Frame CRC will be computed by the CM and added to the frame. Its value is not user-defined.

2.4.5 Packetizing IVN messages

To optimize the performance of the communication, it is possible to limit and reuse the first header section (from the field Device ID to the field Device Flags) only once over multiple IVN Data. As introduced in Section 2.4.1, “TECMP Header”, the solution consists in transferring at once several IVN messages packetized in a single TECMP frame. The principle applies for both the transfer to the attached sinks (for capturing data) and to the CM (for replaying data). The first 12 Bytes of the TECMP Header are inserted only once, at the beginning of the packetized IVN messages. The remaining header section (16 Bytes, from the field Interface ID to the field Data Flags) is inserted once before each captured IVN message. No padding is used between the successive messages inside the TECMP message.

Figure 13 shows a TECMP frame with a single CAN message contained. Here, packetization is not used.

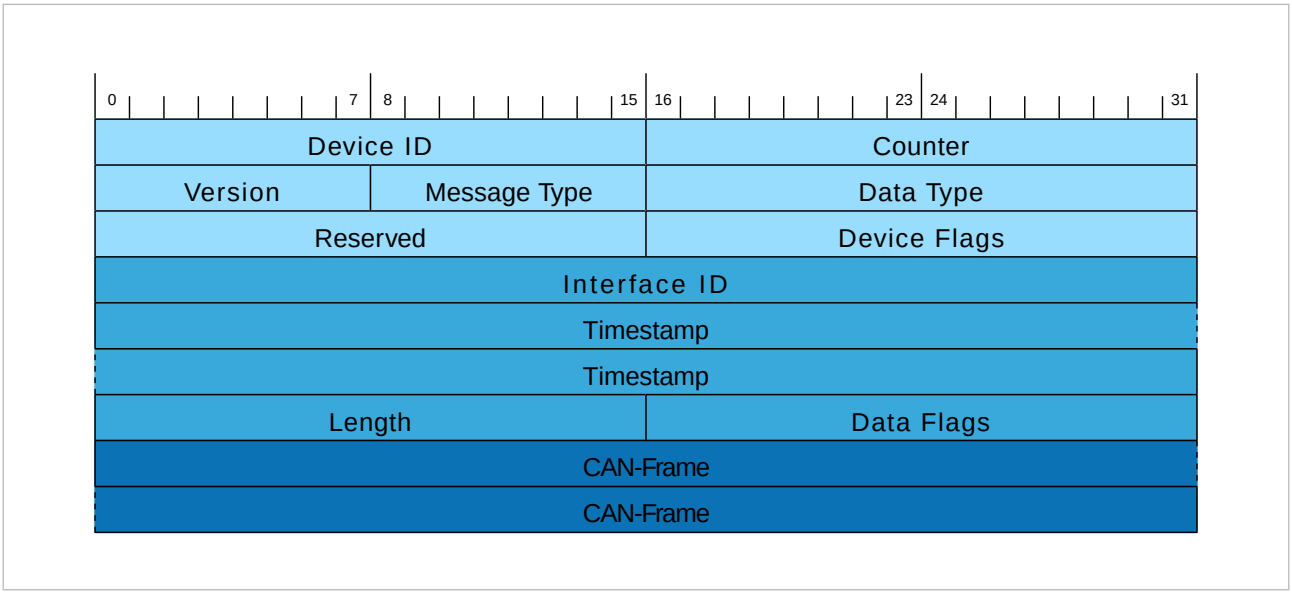


Figure 13 — Transport of a single CAN message without packetization.

By opposition to Figure 13, Figure 14 shows the packetization of four CAN messages into a TECMP frame.

As the identification of the Message Type belongs to the first header section, it applies globally to all the packetized IVN messages. It means that only messages of the same Message Type can be packetized together: Logging stream with Logging Stream or Replay Data with Replay Data. There is no packetization of Control Messages.

Similarly, only messages of the same Data Type (CAN with CAN for instance) can be packetized together.

When capturing data:

- If, despite the packetization of several messages, the minimum size of a 64-Byte Ethernet frame cannot be reached, the CM inserts appropriate zero padding at the end of the TECMP message. There is no padding between the successive messages.
- The protocol does not provide a dedicated field indicating how many messages are packetized because this information can be retrieved based on the value of the length (header field) of the Data and the overall buffered length of the TECMP message.

When replaying data:

- Only FlexRay messages can be packetized. Other protocols are not supported. Successive individual TECMP messages shall be used instead.

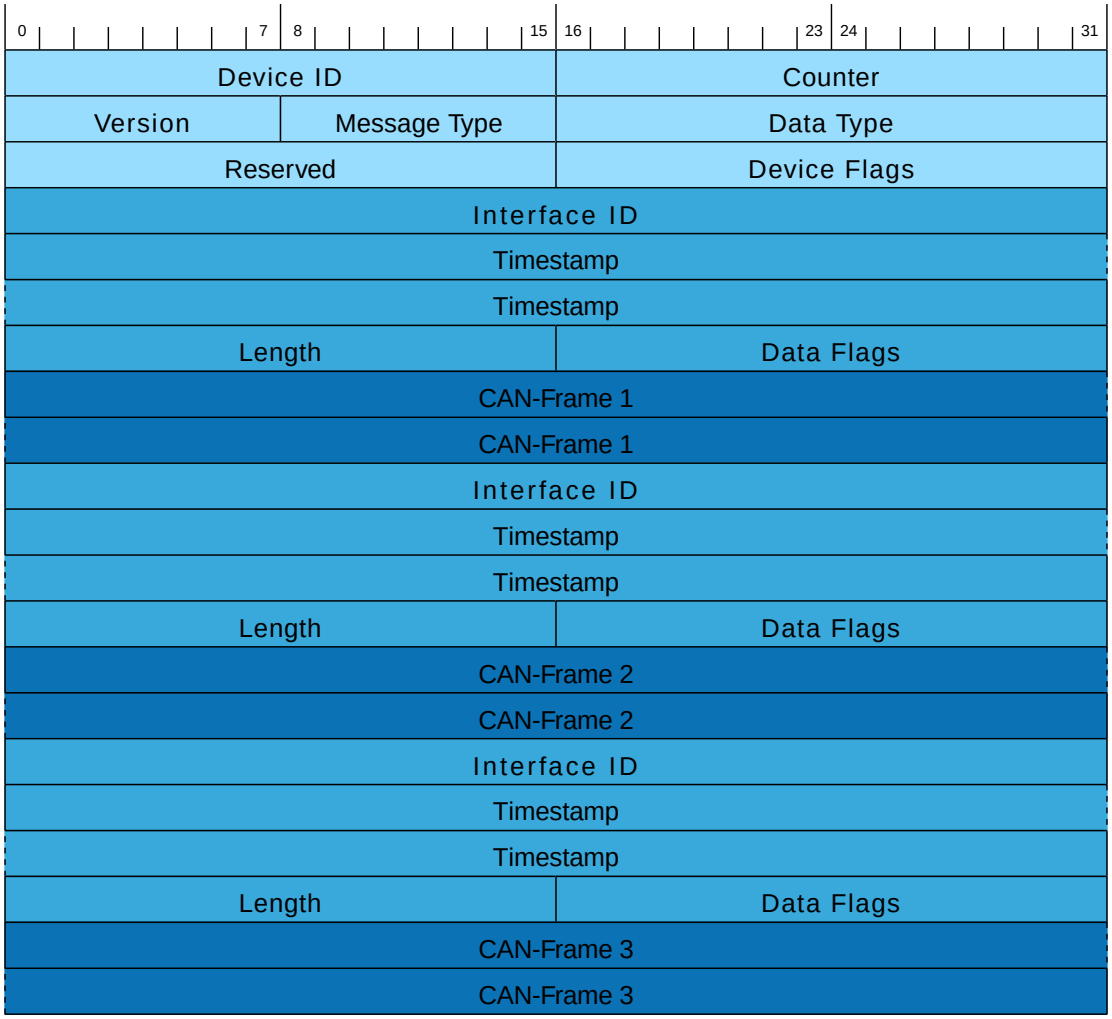


Figure 14 — Transport of a multiple CAN messages with packetization.

2.5 Status Message

Specific Status Messages can be generated by CMs. Those messages are sent in parallel with the Logging Stream and can be used by the sink to identify or debug the status of the capturing interface of the CM or the CM itself during the recording. Additionally, the configuration of the CM during the recording can also be retrieved:

1. Following the occurrence of an internal event (Timeout period), a Status Message is internally generated.
2. This Status Message is sent towards the sink.

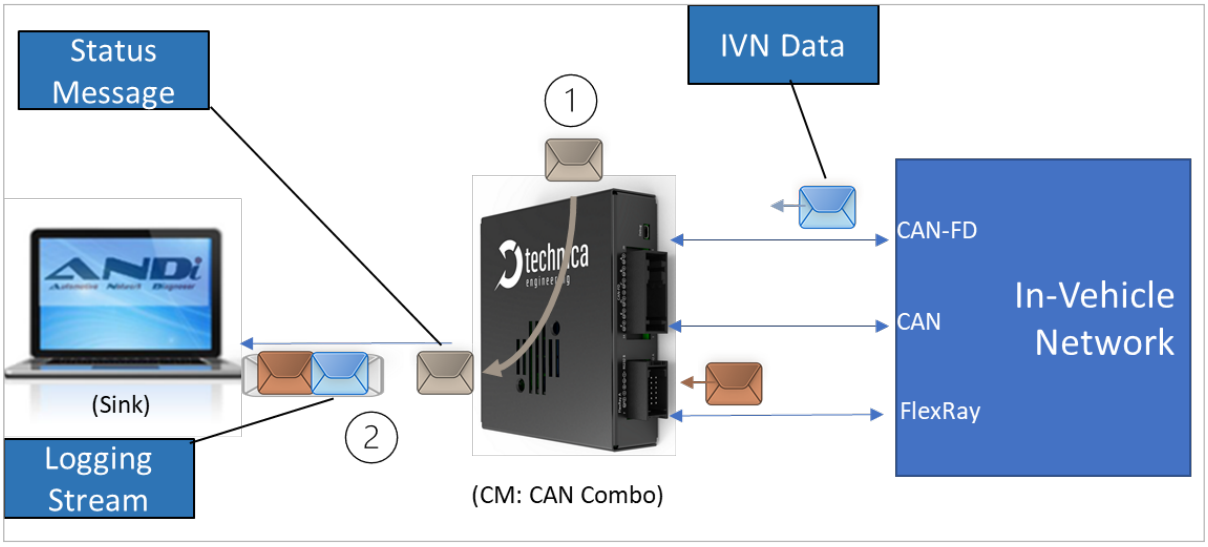


Figure 15 — Status Message generation

Three different Status Messages are currently supported by CMs:

- Status Configuration
- Status Bus/IVN
- Status Device

The Status Messages use the same TECMP Header as other TECMP messages presented in Figure 6. Table 29 shows where the Status Message payload is located at he end of the header.

Device ID	Counter	Version	Message Type	Data Type	Reserved	Device Flags	Interface ID	Time stamp	Length	Data Flags	Status Message payload

Table 29 — Status Messages TECMP overview



As introduced in Table 4, the value of the length field correspond to the length of the entire status message payload.

2.5.1 Status Configuration message

2.5.1.1 Transmission condition

Status Configuration messages are generated cyclically every 60 seconds.

2.5.1.2 Length limitation and Segmentation

Due to the potentially large CM configuration, a dedicated segmentation feature is supported. The maximum length of any Status configuration Message or segmented Status configuration Message is 800 Bytes. If the payload to be transferred does not fit in a single Message, the payload shall be segmented and transferred in multiple and successive Status Messages, each carrying one segment.

2.5.1.3 Format

The payload is presented in the next Tables.

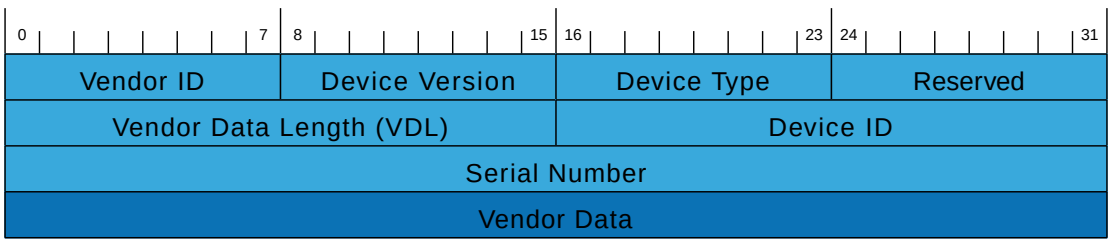


Figure 16 — Status Configuration message payload

Field Name	Length	Description
Vendor ID	1 Byte	Vendor ID - Technica Engineering: 0x0C
Device Version	1 Byte	Versioning of the device
Device Type	1 Byte	Type of device (e.g.: CAN, FR, Ethernet, LIN) , see Table 45
Reserved	1 Byte	For future extensions
Vendor Data Length	2 Bytes	Length of the “Vendor Data” field (in Bytes)
Device ID	2 Bytes	Configured device ID
Serial Number	4 Bytes	Serial number of the device
Vendor Data	n Byte(s)	Technica Engineering specific device configuration. The length n of this field is given by the value of the Vendor Data Length field.

Table 30 — Status Configuration message message fields description

Any Capture Module HW variant can be easily identified based on the notified Serial Number and Capture Module version.

Device Version:
Version of the Capture Module. Used to distinguish different Capture Module variants / revisions.

Vendor ID:

Technica Engineering	0x0C

Device Type:
See Table 45.

Capture Module Serial Number:
This field is 4 Bytes long. In case that the correct serial number for the device is not saved in the memory, the last 4 Bytes of the MAC address will be shown as serial number

Vendor Data:

Multiple configuration parameters are exchanged with the CM in the form of a serialized configuration file. The format of the Vendor Data field is depicted and described in the next two illustrations.

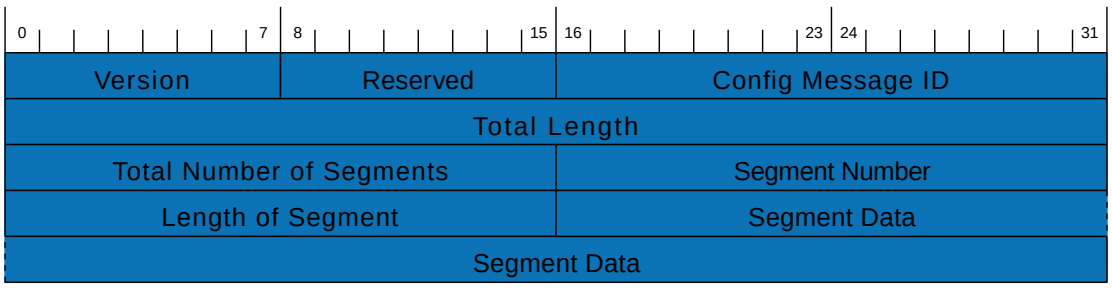


Figure 17 — Vendor Data overview

Field Name	Data Type	Description
Version	UINT8	Version number of the Status Configuration message vendor data format
Reserved	UINT8	Reserved field: 0x00
Config Message ID	UINT16	Identifier incremented by one at every new cyclic transmission
Total Length	UINT32	Sum of the lengths of the individual segments
Total number of segments	UINT16	Number of total Status Configuration message segments generated (for this message cycle).
Segment number	UINT16	Identification of the present segment: between 0 and the value of the field “Total number of segments”-1
Length of segment	UINT16	Length (x) of the following “Segment Data” field in Bytes
Segment Data	x * UINT8	Serialized configuration

Table 31 — Status Configuration message Vendor Data fields description

2.5.2 Status Bus/IVN message

2.5.2.1 Transmission condition

Status Bus/IVN messages are generated cyclically every second. CMs with multiple IVN interfaces send the status of all active IVN interfaces.

2.5.2.2 Length limitation

The maximum length is 1400 Bytes.

2.5.2.3 Format

The payload is depicted in the next illustration.

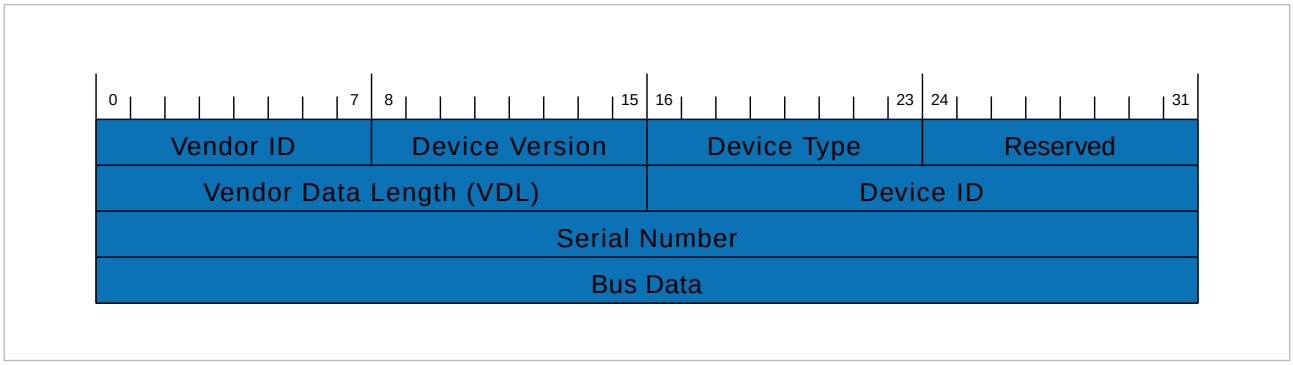


Figure 18 — Status Bus/IVN message payload overview

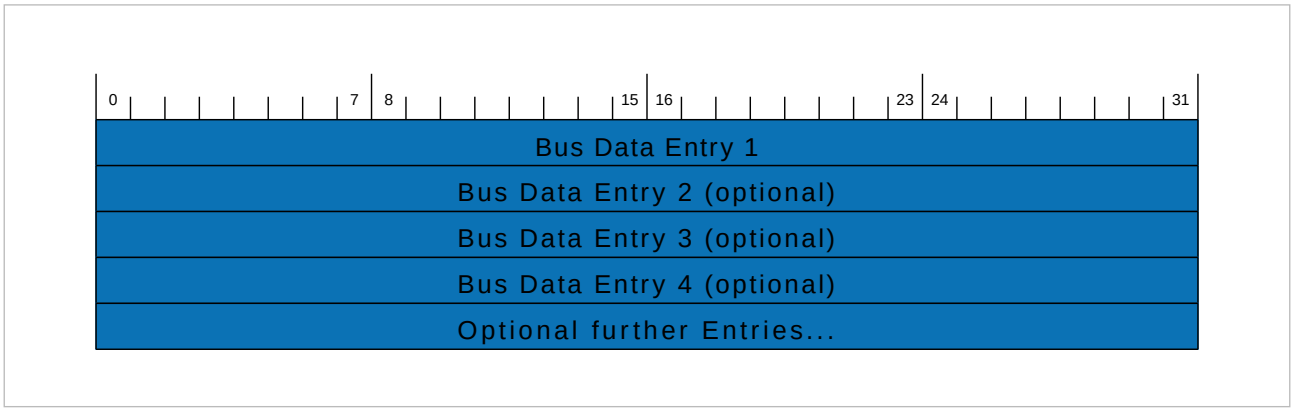


Figure 19 — Bus Data

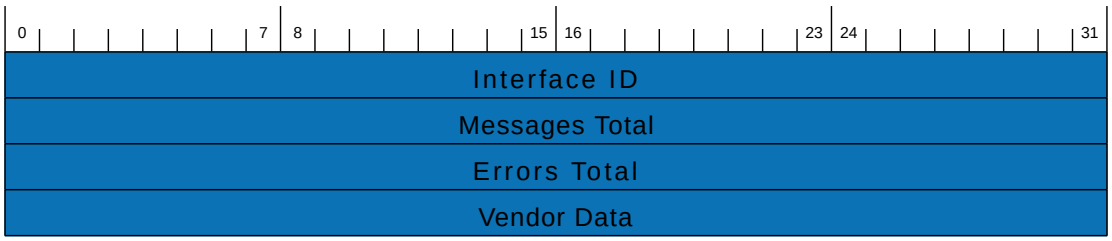


Figure 20 — Bus Data Entry

Field Name	Length	Description
Vendor ID	1 Byte	Vendor ID - Technica Engineering: 0x0C
Device Version	1 Byte	Versioning of the device
Device Type	1 Byte	Type of device (e.g.: CAN, FR, Ethernet, LIN)
Reserved	1 Byte	For future extensions
Vendor Data Length	2 Bytes	Length of the “Vendor Data” field per Bus (in Bytes)
Device ID	2 Bytes	Configured device ID
Serial Number	4 Bytes	Serial number of the device.
Interface ID	4 Bytes	ID that uniquely identifies the log data / bus / link of the IVN on which the Messages/frames were received or sent
Messages Total	4 Bytes	Messages received on the bus/IVN (Interface ID) since the startup of the CM
Errors Total	4 Bytes	Erroneous messages received on the bus/IVN (Interface ID) since the start of the Capture Module
Vendor Data	n Bytes	See description below

Table 32 — Description of the Status Bus/IVN message payload fields

Table 33 respectively Table 34 shows an overview for one respectively multiple bus/IVN interfaces:

Device ID	COUNTER	VERSION	Message Type	DATA TYPE	RESERVED	Device Flags	Interface ID	TIMESTAMP	LENGTH	Data Flags
VENDOR ID	Device VERSION	Device TYPE	RESERVED	LENGTH	Device ID	Serial Number	Interface ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA

Table 33 — Packetizing a single Status Bus/IVN in one Status Bus/IVN message

Device ID	COUNTER	VERSION	Message Type	DATA TYPE	RESERVED	Device Flags	Interface ID	TIME STAMP	LENGTH	Data Flags
VENDOR ID	Device VERSION	Device TYPE	RESERVED	VENDOR DATA LENGTH	Device ID	Serial Number	Interface ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA
							Interface ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA
							Interface ID	MESSAGEs TOTAL	ERRORs TOTAL	VENDOR DATA

Table 34 — Packetizing multiple Status Bus/IVN in one Status Bus/IVN message

Messages Total and Errors Total:
The value of the absolute received messages and respectively messages with errors are available in the respective fields. An error increases both the Messages Total and the Errors Total value. Further information can be transmitted in the Vendor Data.

Vendor Data

The Vendor Data field is used to convey specific information in every Capture Module variant. Consequently, the field has different meaning in different Capture Module types. As an example, for the CM Ethernet Combo, CM 100 High and CM 1000 High variants:

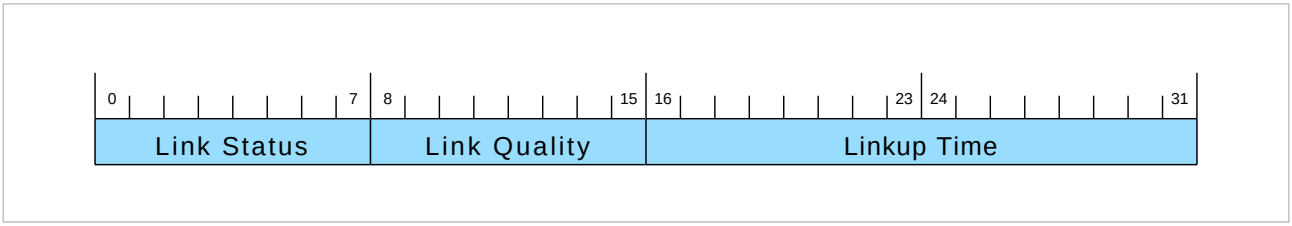
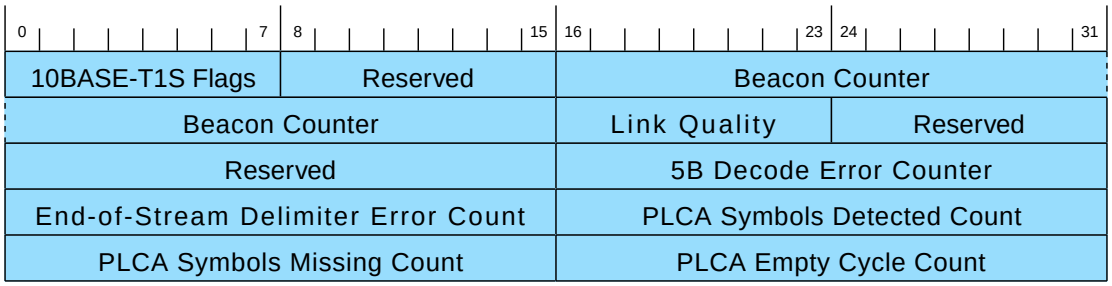


Figure 21 — Status Bus/IVN message Vendor Data format for the CM Ethernet Combo, CM 100 High and CM 1000 High variants.

Field Name	Data Type	Description
Link Status	UINT8	Flag to indicate whether there is Link-Up or not in the corresponding channel during generation of the status message: x00: No x01: Yes
Link Quality	UINT8	Quality of the link of the corresponding channel during generation of the status message, from 0 to 5: 0x00: Link-Down 0x01: Lowest Link quality … 0x05: Highest Link quality
Linkup Time	UINT16	Time (in ms) measured between power up of the CM and link up on this port. Three default reference values are defined based on an internal reference timeout elapsing 1000 ms after power up. • 0x0000: No linkup detected, and the reference timeout did not elapse yet. • 0xFFFE: Invalid: The link was reset after Linkup Time measurement. • 0xFFFF: No linkup detected, and the reference timeout already elapsed. For instance: 0x153 means 339 ms. ⓘ This value is measured only once (at startup). Disconnecting and reconnecting the cable again will result in the value 0xFFFE being reported.

Table 35 — Link Status, Link Quality and Linkup Time fields description.

2.5.2.4 Vendor Data 10BASE-T1S



Field Name	Data Type	Description
10BASE-T1S Flags	UINT8	Flags to indicate general information about 10BASE-T1S: **Offset** / **Name** / **Description** 0 / / 1 / / 2-7 / Reserved / Reserved for future use
Reserved	UINT8	Reserved for future use
Beacon Counter	UINT32	Number of Beacons received on the bus since the startup of the CM. The counter restarts from 0 after overflow.
Link Quality	UINT8	Quality of the link of the corresponding channel, from 0 to 5: 0x00: Unacceptable or Down (0/5). 0x01: Poor (1/5). 0x02: Marginal (2/5). 0x03: Good (3/5). 0x04: Very good (4/5). 0x05: Excellent (5/5).
Reserved	UINT8	Reserved for future use
Reserved	UINT16	Reserved for future use
5B Decode Error Counter	UINT16	The value of the CM local 5B decode error counter. This counter is incremented once whenever the 5B decoder encountered at least one unknown or reserved 5B codeword that could not be decoded, since the last increment. The counter restarts from 0 after overflow.

End-of-Stream Delimiter Error Counter	UINT16	Value of the CM local End-of-Stream Delimiter Error counter. This counter is incremented once whenever at least one End-of-Stream Delimiter Error (ESDERR) or End-of-Stream Jabber (ESDJAB) symbol is detected, since the last increment. The counter restarts from 0 after overflow.
PLCA Symbols Detected Counter	UINT16	The value of the CM local PLCA Symbols Detected counter. This counter is incremented once whenever at least one PLCA BEACON symbol (operating with PLCA disabled) is detected when previously the BEACONs were missing on the channel, since the last increment. The transition from “no BEACON detected” to “BEACON detected” will be counted. The counter restarts from 0 after overflow.
PLCA Symbols Missing Counter	UINT16	Value of the CM local PLCA Symbols Missing counter. This counter is incremented once whenever at least one PLCA BEACON symbol (operating with PLCA disabled) is missing when previously the BEACONs were present on the channel, since the last increment. The transition from “BEACON detected” to “no BEACON detected” will be counted. The counter restarts from 0 after overflow.
PLCA Empty Cycle Counter	UINT16	The value of the CM local PLCA Empty Cycle Status counter. This counter is incremented once whenever at least one empty PLCA bus cycle is detected since the last increment. An empty bus cycle occurs when the node detects no transmissions in any of the possible transmit opportunities between two successive BEACONs. The counter restarts from 0 after overflow.

2.5.2.5 Structure and Vendor Data for CM SerDes

Status Bus for SerDes devices will carry multiple interfaces per SerDes port in this order:

- I2C_1
- I2C_2
- GPIO
- Virtual Channel 1
- Virtual Channel 2
- Virtual Channel 3
- Virtual Channel 4

This is repeated per SerDes interface of the Capture Module. With four SerDes ports you will have 28 interfaces.

Vendor Data definition for I2C data type

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>Bus Errors</td><td>UINT8</td><td>Errors detected on I2C bus:
<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>No ACK error</td></tr></table></td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>For future extensions</td></tr>
</table>

Vendor Data definition for GPIO data type

Field Name	Data Type	Description
Reserved	UINT8	For future extensions
Reserved	UINT8	For future extensions

Vendor Data for Virtual Channel 1 – 4:

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>Bus Errors</td><td>UINT8</td><td><table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>CRC error</td></tr><tr><td>1</td><td>ECC 1-bit error</td></tr><tr><td>2</td><td>ECC 2-bit error</td></tr></table></td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>For future extensions</td></tr>
</table>

2.5.3 Status Device message

2.5.3.1 Transmission condition

Status Device messages are generated cyclically every second.

2.5.3.2 Length limitation

The maximum length is 1400 Bytes.

2.5.3.3 Format

The payload is depicted in Figure 22 and described in Table 36.

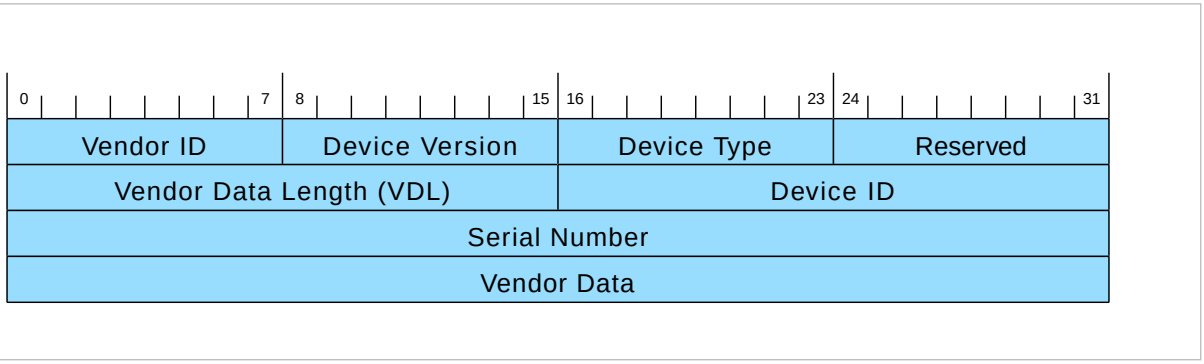


Figure 22 — Status Device Message payload overview

Field Name	Length	Description
Vendor ID	1 Byte	Vendor ID – Technica Engineering: 0x0C
Device Version	1 Byte	Versioning of the device.
Device Type	1 Byte	Type of Capture Module (e.g.: CAN-Combo
Reserved	1 Byte	For future extensions
Vendor Data Length	2 Bytes	Length of the “Vendor Data” field (in Bytes)
Device ID	2 Bytes	Configured ID of the device.
Serial Number	4 Bytes	Serial number of the device.
Vendor Data	n Byte(s)	Specific status information (e.g.: temperature processor, or device’s error conditions)

Table 36 — Description of the Status device message payload fields

Vendor Data – for all device variants:

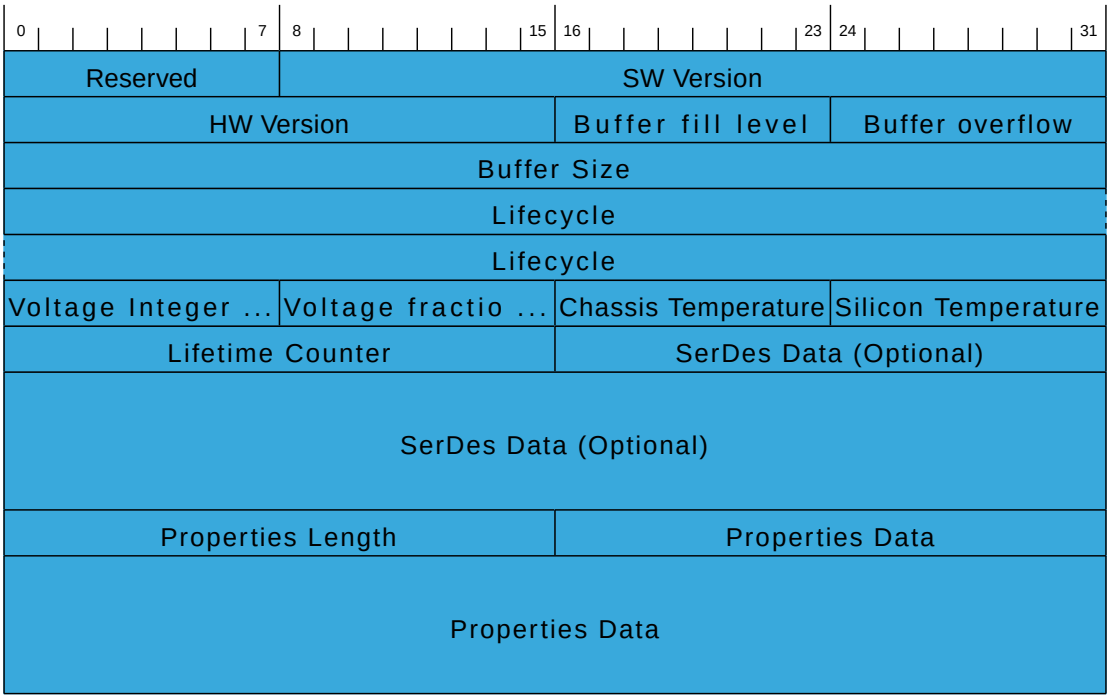


Figure 23 — Status Device message Vendor Data overview

Field Name	Data Type	Description
Reserved	UINT8	Reserved
SW Version	3x UINT8	vX.Y.Z, one Byte for each part. E.g. v8.6.40 → 0x08 0x06 0x28
HW Version	2x UINT8	vX.Y, one Byte for each part. First Byte is represented as decimal Second Byte is represented as hexadecimal E.g. 0x03.0x1A -> v3.1a
Buffer fill level	UINT8	Percentage of internal memory for reception currently used. Possible values: 0, 10, 20, 30, 40, 50, 60, 70, 80, 85, 90, 95. • For the Rx Buffer, 100% is not possible because the memory is emptied before it reaches that level. • A value of 0% can also indicate that no memory is available.
Buffer overflow	UINT8	Flag to indicate whether the memory reached its upper internal memory limit while storing input packets from IVN/Bus inside the internal memory of the CM. x00: No, x01: Yes In case this limit is reached, the CM takes internal countermeasures, stopping recording until the internal memory returns to a functioning level. Once recording is restarted, the flag is reset. Any message received during this time was lost.
Buffer size	UINT32	Size of the internal memory in Gigabits.
Lifecycle	UINT64	Time since startup, in ns.
Voltage integer part	UINT8	Voltage value, the part before the comma. E.g.: 12,30 V → 12 = 0x0C
Voltage fractional part	UINT8	Voltage value, the part after the comma. E.g.: 12,30 V → 30 = 0x1E
Chassis Temperature	SINT8	Temperature of the CM chassis in °C. • 0x00 up to 0x7E → 0°C up to 126°C • 0x7F → temp ≥ 127°C • 0x80 → Not Available • 0xFF down to 0x81 → -1°C down to -127°C

Field Name	Data Type	Description
Silicon Temperature	SINT8	Internal temperature (junction temperature) of the CM processing unit in °C. • 0x00 up to 0x7E → 0°C up to 126°C • 0x7F → temp ≥ 127°C • 0x80 → Not Available • 0xFF down to 0x81 → -1°C down to -127°C
Lifetime Counter	UINT16	Hour counter. • 0xFFFF → Not Available
SerDes Data	UNIT8[34]	See Section 2.5.3.4, "Vendor Data SerDes" for format Only present for SerDes devices, see Section 2.7.5, "Device Type" for applicable devices
Properties Length	UINT16	Length of the Properties Data field

Properties Data	UINT8[N]	Device properties as UTF-8 string. One line per property (separated by '\n' character). Each property has a key and value separated by '=' character.
		Example:
		`fpga.release=R013.003.000`
		`boot.release=R013.003.000`

Table 37 — Status Device message Vendor Data fields description

2.5.3.4 Vendor Data SerDes

Field Name	Data Type	Description
Initialization Error Flags	UINT16	Last 4 bits indicate status for SerDes initialization sequence.
SFP+ A TX Frames	UINT64	Counter of transmitted frames on SFP+ A.
SFP+ B TX Frames	UINT64	Counter of transmitted frames on SFP+ B.
SFP+ C TX Frames	UINT64	Counter of transmitted frames on SFP+ C.
SFP+ D TX Frames	UINT64	Counter of transmitted frames on SFP+ D.

The Field "Initialization Error Flags" is defined as follows:

Bit position	15 – 4	3	2	1	0
Description	RESERVED	Initialization Error on Port 4: 0: no error 1: error	Initialization Error on Port 3: 0: no error 1: error	Initialization Error on Port 2: 0: no error 1: error	Initialization Error on Port 1: 0: no error 1: error

2.6 Control Message

To control the recording system and all possible connected systems on the logging network (consisting of several loggers, Captures Module and instrumentation adapters) and signaling events in the recorded trace, specific Control messages are defined by the TECMP protocol. Which Control Message is active in the measurement components depends on their configuration. All CMs can be informed about the occurrence of a configurable Control message. Received Control messages are passed on by all components inside the measurement network. Control Messages are either received by a CM (Figure 24) or generated by it (Figure 25).

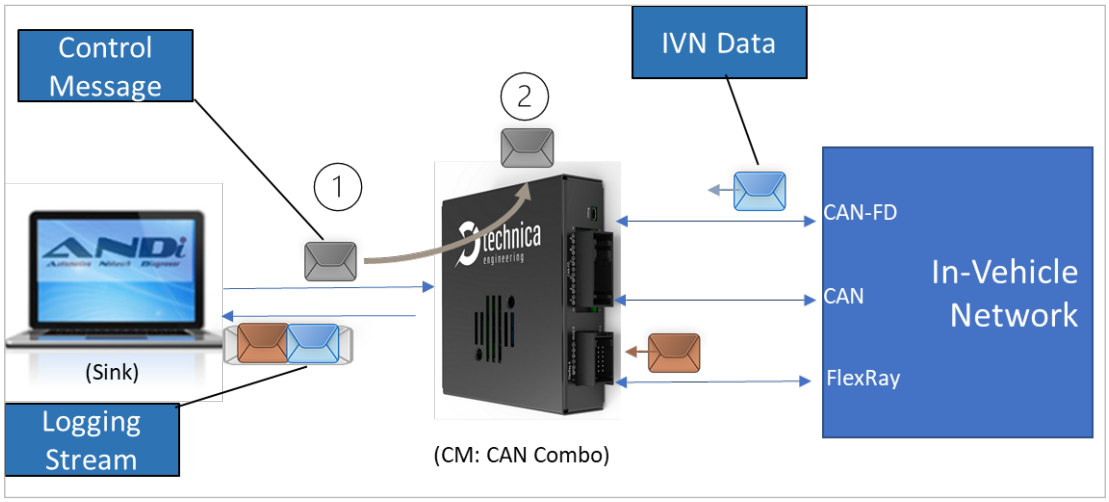


Figure 24 — Reception of a Control Message by the CM.

In Figure 24:

1. The CM receives a Control Message.
2. It is internally checked and processed according to the Control Message ID (Table 44).

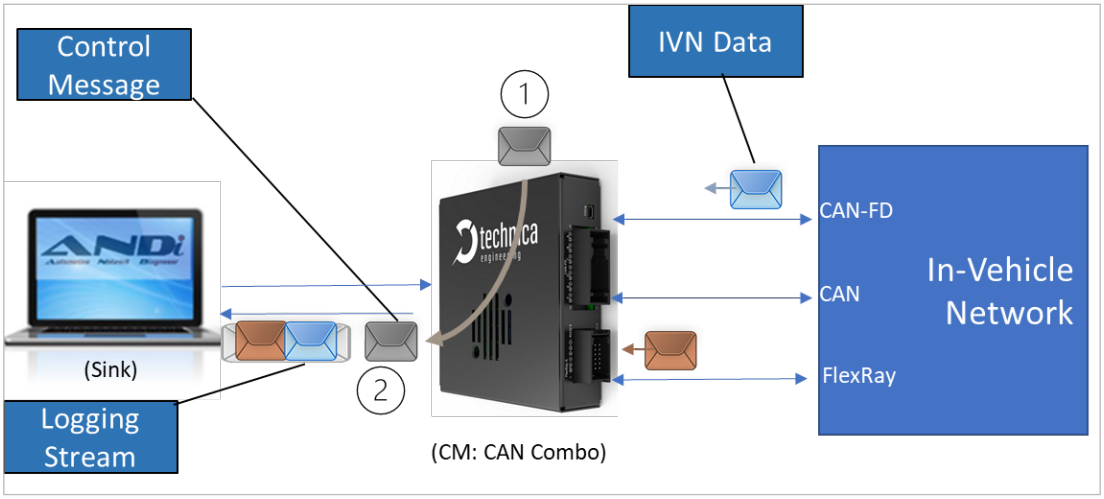


Figure 25 — Generation of a Control Message by the CM.

In Figure 25:

1. Following the occurrence of an event (Ethernet Link down detected, time synchronization acquired…), a Control Message is internally generated.
2. This Control Message is sent towards the sink.

The Control Messages use the same TECMP Header as other TECMP messages presented in Figure 6. Figure 26 shows where the Control Message payload is located at he end of the header.

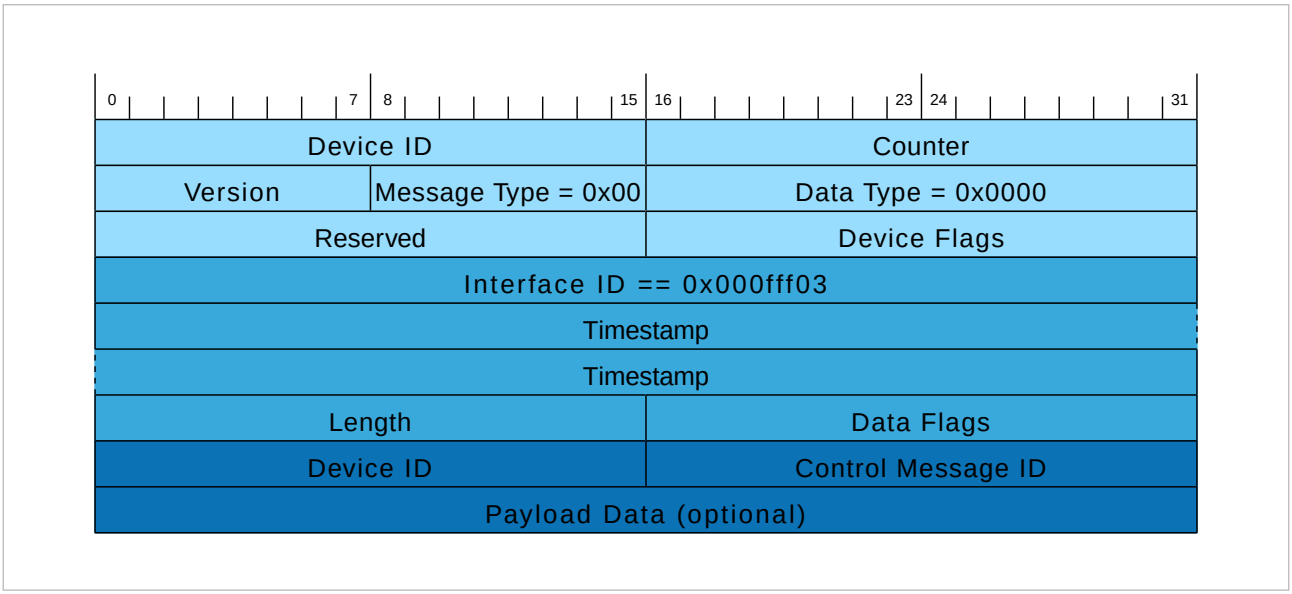


Figure 26 — Control Messages TECMP overview



As introduced in Table 4, the value of the length field correspond to the length of the entire control message payload: Device ID, Control Message ID and the optional Payload Data field.

Field Name	Length	Description
Device ID	UINT16	Identifies the sender of the Control Message (see Table 43).
Control Message ID	UINT16	Identifier of the specific Control Message (see Table 44).
Payload Data	UINT8[x]	Optional additional information, whose length can be computed based on the value of the Length field.

Table 38 — Control message fields description

2.6.1 CAN Replay Fill Level

The CAN Replay fill level is a special Control Message, supporting the CAN Replay feature:

1. The CAN(-FD) data to transmit on the IVN is received by the CM in the form of one or more TECMP messages.
2. The CAN(-FD) data to transmit on the IVN is sent out by the CM accordingly to the CAN Arbitration principle.
3. Depending on the Busload and the replaying scheme, it might happen that the memory resource of the CM is put under too much pressure. This case coincides with the generation of the CAN Replay fill level Control Message.
4. This Control Message is sent to the Sink.

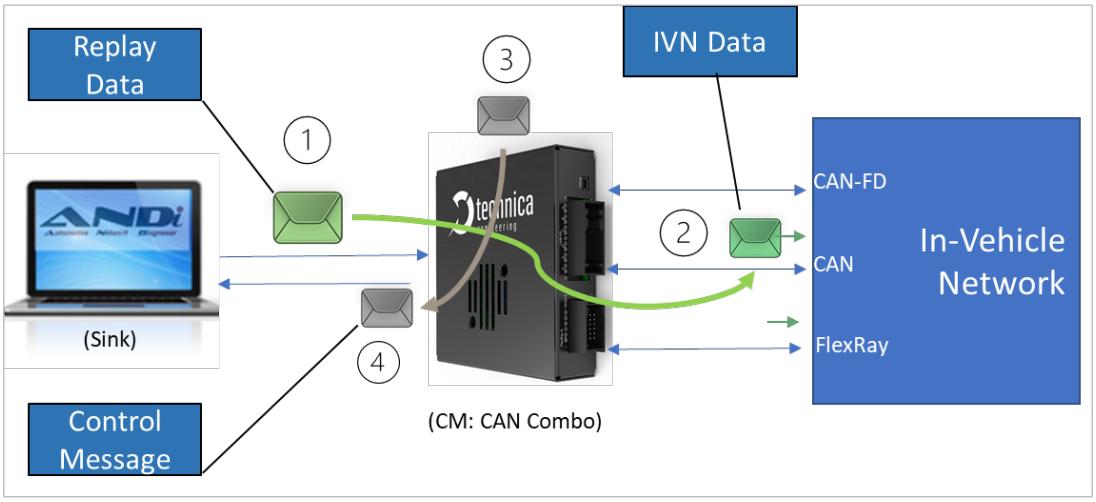


Figure 27 — Generation of the CAN Replay fill level Control Message

The Payload Data format of the CAN Replay fill level is defined as follows:

Name	Length	Description
DDR fill Level	UINT8	Percentage of the DDR memory currently used. Possible values: 0, 10, 20, 30, 40, 50, 60, 70, 80, 85, 90, 95, 100.
DDR Buffer Overflow	UINT8	Flag to indicate whether there is a memory overflow or not. x00: No, x01: Yes
CAN Queue size	UINT8	Maximum number of frames which can be stored in one CAN Queue.
CAN Queue fill level	UINT8[x]	Number of frames currently stored in the Queue. x: depends on the number of CAN(-FD) interfaces of the PLP-enabled-device: • Number of CAN(-FD) interfaces.

Table 39 — CAN Replay fill level

2.6.2 FlexRay POC State

This Control Message is generated by the CM towards the Sink upon POC transition of an active FlexRay Interface.

Name	Length	Description
Interface ID	UINT32	ID of the FlexRay Interface.
State	UINT8	0x0: CONFIG 0x1: DEFAULT_CONFIG 0x2: HALT 0x3: NORMAL_ACTIVE 0x4: NORMAL_PASSIVE 0x5: READY 0x6: STARTUP 0x7: WAKEUP Else: unknown

2.6.3 10BASE-T1S

Configurable parameters on the web:

- Events/errors that will trigger the Control Message on a particular interface
- Resolution for the event triggered Control Message
  - Limited to avoid that the trace is continuously full of errors/events
  - Control Message is event triggered with a polling/debouncing time > 20 ms.

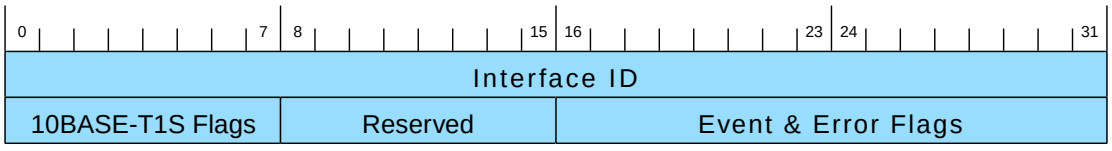


Figure 28 — Control Messages Vendor Data for 10BT1S variant (configured by Web Interface)

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>Interface ID</td><td>UINT32</td><td>Interface ID of the interface where the Event/errors occurred</td></tr>
<tr><td>10BASE-T1S Flags</td><td>UINT8</td><td>Flags to indicate general information about 10BASE-T1S:
<table>
<tr><th>Offset</th><th>Name</th><th>Description</th></tr>
<tr><td>0</td><td>Beacons received</td><td></td></tr>
<tr><td>1</td><td>PLCA Enabled</td><td></td></tr>
<tr><td>2 - 7</td><td>Reserved</td><td>Reserved for future use.</td></tr>
</table></td></tr>
<tr><td>Reserved</td><td>UINT8</td><td>Reserved for future use.</td></tr>
<tr><td>Event & Error Flags</td><td>UINT16</td><td>Flags to indicate some errors or events that could be reported by the CM-10BASE-T1S about this interface.

Flags will be set for all event/errors that occurred since the event/error occurrence that triggered the Control Message.
<table>
<tr><th>Offset</th><th>Name</th><th>Description</th></tr>
<tr><td>0</td><td>5B Decode Error</td><td></td></tr>
<tr><td>1</td><td>End-of-Stream Delimiter Error</td><td></td></tr>
<tr><td>2</td><td>PLCA Symbols Detected</td><td></td></tr>
<tr><td>3</td><td>PLCA Symbols Missing</td><td></td></tr>
<tr><td>4</td><td>PLCA Empty Cycle</td><td></td></tr>
<tr><td>5 - 15</td><td>Reserved</td><td>Reserved for future use.</td></tr>
</table></td></tr>
</table>

2.6.4 OPEN TC10 Sleep/Wake-Up

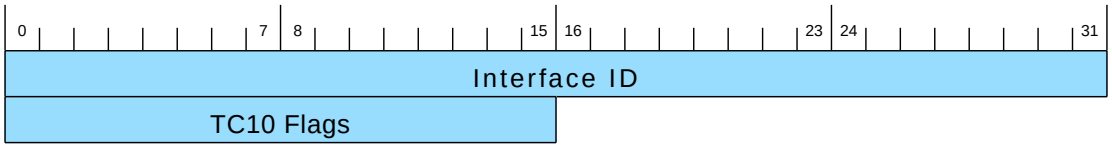


Figure 29 — Overview of a TC10 events

<table>
<tr><th>Field Name</th><th>Data Type</th><th>Description</th></tr>
<tr><td>Interface ID</td><td>UINT32</td><td>Interface ID of the interface where the Event/errors occurred</td></tr>
<tr><td>TC10 Flags</td><td>UINT16</td><td>Flags to indicate general information about TC10:
<table>
<tr><th>Offset</th><th>Name</th><th>Description</th></tr>
<tr><td>0</td><td>WAKEUP</td><td>‘1’: local or remote wake-up detected
‘0’: no : local or remote wake-up detected</td></tr>
<tr><td>1</td><td>LPS_RECEIVED</td><td>‘1’: LPS code groups received
‘0’: NO LPS code groups received</td></tr>
<tr><td>2</td><td>LINK_UP</td><td>‘1’: link UP event received
‘0’: NO link UP event received</td></tr>
<tr><td>3</td><td>LINK_DOWN</td><td>‘1’: link DOWN event received
‘0’: NO link DOWN event received</td></tr>
<tr><td>4</td><td>SLEEP_ABORT</td><td>‘1’: transition from Sleep Request back to Normal mode as a result of a Sleep Request expiring
‘0’: no transition from Sleep Request back to Normal as a result of the Sleep Request timer expiring</td></tr>
<tr><td>5 - 15</td><td>Reserved</td><td>Reserved for future use.</td></tr>
</table></td></tr>
</table>

Table 40 — Format of TC10 events

2.7 TECMP fields and IDs

The supported values of the TECMP header fields are documented in this chapter.

2.7.1 Message Type

Name	Dec.	Hex.
Control Message	0	0x00
Status Device	1	0x01
Status Bus/IVN	2	0x02
Logging Stream	3	0x03
Status Configuration	4	0x04
Replay Data	10	0x0A

Table 41 — Message Type



“Logging Stream” identifies IVN data messages which were received by the CM. “Replay Data” identifies IVN data messages which are sent by the CM.

2.7.2 Data Type

Definition	Dec.	Hex.	Replay?
CAN	2	0x0002	Yes
CAN-FD	3	0x0003	Yes
LIN	4	0x0004	Yes
FlexRay	8	0x0008	Yes
GPIO	10	0x000A	No
ILaS	14	0x000E	No
UART/RS232	16	0x0010	No
Analog	32	0x0020	No
Ethernet II (classical Ethernet frame)	128	0x0080	Direct
Ethernet RAW	129	0x0081	Yes
Ethernet 10BASE-T1S	130	0x0082	Direct
Ethernet 10BASE-T1S PLCA Raw Logging	131	0x0083	Direct
I2C	260	0x0104	No

Table 42 — Data Type



Ethernet II data and Ethernet 10BASE-T1S data can be transmitted by sending the data directly to the Capture Module and do not require a special Replay format.

2.7.3 Device ID

To enforce interoperability, the IP addresses of the CMs is defined based on the IP-network 10.104.3.xxx.

Capture Module	IP Address 10.104.3.xx	Device ID dec. 2 Bytes	Device ID hex. 2 Bytes	Rotary switch position (0x)
LIN Combo				
LIN Combo 0	10.104.3.48	48	0x30	0
LIN Combo 1	10.104.3.49	49	0x31	1
LIN Combo 2	10.104.3.50	50	0x32	2
…	…	…	…	…
LIN Combo 15	10.104.3.63	63	0x3F	F
CAN Combo				
CAN Combo 0	10.104.3.64	64	0x40	0
CAN Combo 1	10.104.3.65	65	0x41	1
CAN Combo 2	10.104.3.66	66	0x42	2
…	…	…	…	…
CAN Combo 15	10.104.3.79	79	0x4F	F

Capture Module	IP Address 10.104.3.xx	Device ID dec. 2 Bytes	Device ID hex. 2 Bytes	Rotary switch position (0x)
Sense				
Sense 0	10.104.3.80	80	0x50	0
Sense 1	10.104.3.81	81	0x51	1
Sense 2	10.104.3.82	82	0x52	2
Sense 3	10.104.3.83	83	0x53	3
Sense 4	10.104.3.84	84	0x54	4
Sense 5	10.104.3.85	85	0x55	5
Sense 6	10.104.3.86	86	0x56	6…E
Sense 7	10.104.3.87	87	0x57	F
100 High				
100 High 0	10.104.3.96	96	0x60	0
100 High 1	10.104.3.97	97	0x61	1
100 High 2	10.104.3.98	98	0x62	2
…	…	…	…	…
100 High 15	10.104.3.111	111	0x6F	F
10BASE-T1S				
10BASE-T1S 0	10.104.3.112	112	0x70	Rotary switch set to 0, Not permitted in logging mode, delivery status.
10BASE-T1S 1	10.104.3.113	113	0x71	1
10BASE-T1S 2	10.104.3.114	114	0x72	2
10BASE-T1S 3	10.104.3.115	115	0x73	3
10BASE-T1S 4	10.104.3.116	116	0x74	4
10BASE-T1S 5	10.104.3.117	117	0x75	5
10BASE-T1S 6	10.104.3.118	118	0x76	6
10BASE-T1S 7	10.104.3.119	119	0x77	7…D
10BASE-T1S 8	10.104.3.120	120	0x78	E
10BASE-T1S 9_reset	10.104.3.121	121	0x79	F
ILaS Sniffer				
ILaS Sniffer 0	10.104.3.122	122	0x7A	Rotary switch set to 0, Not permitted in logging mode, delivery status.
ILaS Sniffer 1	10.104.3.123	123	0x7B	1
ILaS Sniffer 2	10.104.3.124	124	0x7C	2
ILaS Sniffer 3	10.104.3.125	125	0x7D	3…D
ILaS Sniffer 4	10.104.3.126	126	0x7E	E
ILaS Sniffer 5_reset	10.104.3.127	127	0x7F	F
Eth Combo				

Capture Module	IP Address 10.104.3.xx	Device ID dec. 2 Bytes	Device ID hex. 2 Bytes	Rotary switch position (0x)
Eth Combo 0	10.104.3.128	128	0x80	0
Eth Combo 1	10.104.3.129	129	0x81	1
Eth Combo 2	10.104.3.130	130	0x82	2
…	…	…	…	…
Eth Combo 15	10.104.3.143	143	0x8F	F
1000 High				
1000 High 0	10.104.3.144	144	0x90	0
1000 High 1	10.104.3.145	145	0x91	1
1000 High 2	10.104.3.146	145	0x92	2
…	…	…	…	…
1000 High 15	10.104.3.159	159	0x9F	F
Network Interfacer 10BASE-T1S				
NI 10BASE-T1S 0	10.104.3.184	184	0xB8	
NI 10BASE-T1S 1	10.104.3.185	185	0xB9	
NI 10BASE-T1S 2	10.104.3.186	186	0xBA	
NI 10BASE-T1S 3	10.104.3.187	187	0xBB	
SerDes				
SerDes 0	10.104.3.192	192	0xC0	0
SerDes 1	10.104.3.193	193	0xC1	1
SerDes 2	10.104.3.194	194	0xC2	2
…	…	…	…	…
SerDes 15	10.104.3.207	207	0xCF	F
MultiGigabit				
MultiGigabit 0	10.104.3.224	224	0xE0	0
MultiGigabit 1	10.104.3.225	225	0xE1	1
MultiGigabit 2	10.104.3.226	226	0xE2	2
…	…	…	…	…
MultiGigabit 15	10.104.3.239	239	0xEF	F
Other				
Reserved	n/a	256 – 65535	0x100 – 0xFFFF	n/a

Table 43 — Device ID

2.7.4 Control Message ID

The type of control message is defined via a 2 Bytes long Control Message ID field.

Name	ID	Source	Payload Data	Description
Logger Ready	0x02	Logger	None	This control message notifies the logger readiness to receive data. It shall be sent cyclically every second.
CAN Replay fill level	0xE0	CM	See Section 2.6.1, "CAN Replay Fill Level".	Fill level of the CAN transmission feature.
FlexRay POC State	0xE1	CM	See Section 2.6.2, "FlexRay POC State"	Notification of the current FlexRay POC State.
Status 10BASE-T1S	0xE2	CM	See Section 2.6.3, "10BASE-T1S"	10BASE-T1S Event/Errors
TC10 Events	0xE3	CM	See Section 2.6.4, "OPEN TC10 Sleep/Wake-Up"	TC10 Events
IO Channel Events	0xE4	CM	Application-specific	IO Channel Events
Reserved	0xE5 – 0xFF	n/a	n/a	Reserved for future use.
Linkup detected	User-configurable	CM	None	This control message notifies the occurrence of a Linkup event on the Ethernet link. It is a single-shot message.
Linkdown detection	User-configurable	CM	None	This control message notifies the occurrence of a Link-down event on the Ethernet link. It is a single-shot message.
Time synchronization acquired	User-configurable	CM	None	This control message notifies the acquisition of the time-synchronization. It is a single-shot message.
Time synchronization lost	User-configurable	CM	None	This control message notifies the loss of the time-synchronization. It is a single-shot message.

Table 44 — Control Message ID

2.7.5 Device Type

The following table describes the value of the Device Type (1 Byte) field.

Name	ID
CM LIN Combo	0x02
CM CAN Combo	0x04
CM 100 High	0x06
CM 100 High TC10	0x07
CM Eth Combo	0x08
CM 1000 High	0x0A
CM 10BASE-T1S	0x0C
ILaS Sniffer	0x0E
CM SerDes GMSL2/3	0x40
CM MultiGigabit	0x42
EES	0x44
CM Sense	0x46
CM SerDes ASA ML	0x48
Network Interfacer 10BASE-T1S	0x52
Other	0xFF

Table 45 — Device Type

2.7.6 Interface ID

The following table describes the value of the Interface ID (4 Bytes) field for the Status and Control messages.

Name	ID
Status Bus/IVN message	0x000FFF00
Status Configuration message	0x000FFF01
Status Device message	0x000FFF02
Control message	0x000FFF03

Table 46 — Interface ID

3 Bibliography

- IEEE Std 802.3-2022 - IEEE Standard for Ethernet

4 Contact

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5 Changelog

Version	Chapter/Section	Description	Date
1.0	All	First release.	10.02.2020
1.1	2.4 and 2.6.1	References to “User Event” replaced with “Control Message”.	12.02.2020
1.2	2.3.2, 2.3.3, 2.6.3 2.2.2.5 2.6.1 2.3	Description of the fields of Vendor Data (Status Bus/IVN messages and Status Capture Module Message) added. CM IP-address list generalized. References to “Length of vendor data” replaced with “Vendor Data Length” (VDL). Added value description to the TECMP FlexRay Data flags. Added “Replay Data” Message type and frame format description. Aligned cyclic transmission of Status Message.	06.04.2020
1.3	2.6.3 2.6.5 2.3 2.3.2 2.3.1	Device ID table (switch position) updated with hexadecimal value. Capture Module naming updated. Cyclic time of Status Messages updated. Added upper bound length of 1400 Bytes. Added Linkup time to Status bus/IVN message payload. Added description of Status Configuration message vendor data. Added 1000 High Device ID.	27.05.2020
1.4	2.2.1 2.2.2.4 2.2.2.5	Added restriction on Timestamp. “CRC Error” Data Flag bit detailed for FlexRay. “Tx” Data Flag bit detailed. Added “BRS” bit to the CAN-FD Data frame. Added “CAS” bit to the FlexRay Data frame.	25.08.2020
1.5	1.1 – 2.1 2.3.1 2.3.3	Editorial changes. Status Configuration message cycle time updated from 1s to 60s. Field “Temperature” renamed to “Chassis Temperature”. Update from UINT8 to SINT8. Added “Silicon Temperature” field.	13.11.2020

Version	Chapter/Section	Description	Date
1.6	Whole document	The major following changes were performed: • Added new “Replay Data” feature. • “Channel ID” renamed to “Interface ID”. “CM ID” renamed to “Device ID”. • Logging feature extended: Source Data and Data Flags updated for CAN(-FD) and FlexRay.	13.12.2021
1.6.1	2.4.3.3	Fixed LIN Data Flags section (bits 4-3).	15.12.2021
1.7	Whole document	Numerous changes, including the following: Removed wrong reference to “serialized JSON”. Added new FlexRay POC Status Control Message Description of value 0 for “No Slave Response” in LIN Rx corrected. Rework (Maximum length of Status Configuration Message updated from 1400 to 800 Bytes, Segmentation) Extended functionalities for CAN(-FD) Rx and Tx, LIN Rx and Tx. Removed “Threshold” field for Analog Rx. Data Type of Samples corrected from UINT to INT. Size of the “Unit” field updated from two to three bits. Corrected timeout for “clock asynchronous” from 250ms to 375ms. “Source Data” renamed to Data. Fields containing “CM” or “Capture Module” renamed with “Device”. Status Capture Module message renamed to Status Device message. Description of HW Version in Status Device Message updated. Data Type and Control Message ID summary table updated.	29.07.2022
1.8	Whole document	Numerous changes, including the following: • Include TECMP 10Base-T1s v2.2 • Include TECMP ILaS Sniffer • Include Device and Type ID MultiGig • Corrected checksum info on MsgType = 0x0004 • Corrected interface ID status config/Status Bus • Corrected Control Message CAN fill level • Add analog sample values (less than 1ms)	n/a

Version	Chapter/Section	Description	Date
1.9	Whole document	• 10BASE-T1S update. • Updating Device Names (ILaS Sniffer, Multigigabit, Sense). • Remove “unused” from Replay specification table. • Added CM 100 High TC10 device type. • Updated data types in Table 42. • Added I2C and GPIO for CM SerDes. • Added Status Bus and Status Device format for CM SerDes	28.04.2025
1.9.1	2.5.2.3 2.6	• Add 0xFFFE to Linkup Time values • Fix Length in control message fields description	16.09.2025
1.10.0	Whole document	• Added RTR flag to CAN Tx Data Flags. • Added RST_LIN_TABLE flag to LIN Tx Data Flags. • Added TC10 Events Control Message. • Added Device Properties String to Status Device message. • Fixed Vendor Data 10BASE-T1S data types. • Added FlexRay Channel flag to FlexRay Tx Data Flags. • Added PLCA Symbol Logging data type (0x0083) for 10BASE-T1S. • Renamed Abbreviations section to Glossary; expanded entries. • Fixed checksum documentation by removing incorrect upper bit note. • Reserved Control Message ID 0xE4 for IO Channel Events.	17.07.2026